

HYPOSAT 6.3 and HYPOMOD 2.3 – The User Manual

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1 INTRODUCTION

HYPOSAT is a program developed to locate single seismic sources and **HYPOMOD** is a program to calculate residuals of observed data for a given seismic source location. Both programs utilize observed onset time, backazimuth (*i.e.*, station-to-event azimuth, BAZ) and ray-parameter observations. Additionally, all possible travel-time differences between different onsets at individual stations can be included in the location process (*e.g.*, the onset-time difference between the phases PcP and P ($t_{PcP} - t_P$) gives additional constraint for the source depth). If amplitude and period measurements are available for P or S onsets or for surface waves, individual station and network magnitudes can also be estimated. The files ***HYPOSAT_HISTORY.md*** and ***HYPOMOD_HISTORY.md***, which are part of the program documentation, contain listings of the program versions and main changes from version to version.

For clarity, all file names in this Manual are written in bold *italics*. This document and all **HYPOSAT / HYPOMOD** output are consequently using the phrases ‘location uncertainty’, ‘model uncertainty’ and ‘data uncertainty’ instead of ‘location error’, ‘model error’ and ‘data error’ and it is clearly distinguished between the terms ‘backazimuth’ and ‘azimuth’, where backazimuth (BAZ) is used for defining the direction from the station to the epicenter or from which the seismic wave arrives at the station measured from North clockwise over East in [°], *i.e.*, $0^\circ \geq \text{BAZ} \geq 360^\circ$.

Seismic phases considered in both programs are those included in IASPEI 1991-type travel-time tables. The program follows the seismic phase name convention as recommended by IASPEI on its General Assembly in Sapporo, in July 2003 (Storchak *et al.*, 2002; 2003; 2006; 2009; 2011; Bormann *et al.*, 2013; Schweitzer *et al.*, 2021a). A software package (*libtau*-software) for travel-time-table calculations for spherical Earth models is distributed with the IASPEI 1991 tables. It is based on the tau-spline algorithm (Buland & Chapman, 1983) and was implemented and modified here so that all precalculated data files become platform independent ASCII files. In addition to the original software version, diffracted PKPbc and double surface reflections of body waves (*e.g.*, PPP or S'S'S'), depth phases for surface reflections (pPPP & sSS, pP'P' *etc.*), depth phases of core reflections (pPcP, sScS), single surface reflections of core phases (PcPPcP & ScSScS) as well as S3KS were included in the travel-time-table calculations. Reflections from the Conrad discontinuity – in the following abbreviated as ‘the Conrad’ – (PbP, SbS) and the Mohorovičić discontinuity – in the following abbreviated as ‘the Moho’ – (PmP, SmS), are included if local/regional seismic velocity models are used.

For most of the phases or their proxies, ellipticity corrections are automatically calculated with the adapted software of Kennett & Gudmundsson (1996). Instead of the original Kennett & Gudmundsson coefficients, the program uses the updated and corrected set of coefficients from Russell *et al.* (2022). However, note that for double surface reflections (*e.g.*, PgPgPg, SnSnSn, PPP, SSS, P'P'P' *etc.*), ellipticity corrections cannot be applied at this moment.

Since **HYPOSAT 6 / HYPOMOD 2** (see Schweitzer, 2018), the global crustal model Crust 5.1 (Mooney *et al.*, 1998) with $5^\circ \times 5^\circ$ resolution has been replaced by the global crustal model CRUST 1.0 (Laske *et al.*, 2013) with $1^\circ \times 1^\circ$ resolution. With **HYPOSAT 6.1 / HYPOMOD 2.1**, the usage of CRUST 1.0 is extended: when requested, the program can now correct reflections at

the Earth's surface not only for topography but also for the water-depth of oceans and larger lakes (as far as defined in CRUST 1.0); for further details see Schweitzer (2025). With this change, pwP-type phases are now included, but the program will also correct all types of P- or S-phase reflections at the surface and may change the phase name accordingly. As an extension to the IASPEI phase-name convention (Storchak *et al.*, 2002; 2003; 2006; 2009; 2011; Bormann *et al.*, 2013; Schweitzer *et al.*, 2021a), **HYPOSAT** / **HYPOMOD** allow not only pP changing to pwP but also *e.g.*, PP to PwP or P'P' to P'wP').

HYPOSAT runs in an iterative mode, in which each iteration step is based on the solution of an equation system with the Generalized Matrix Inversion (GMI) (see *e.g.*, Bolt (1970) and Niewiadomski (1989) or the textbooks of Aki & Richards (1980) or Menke (1989)). All partial derivatives for the matrix inversion are recalculated before each iteration. After each iteration, the program evaluates the inversion results and either continues with another iteration step or stops all processing and generates an output with the location results. The user can steer this processing with different options for the usage of the input data and theoretical travel-time models and additional parameters for the inversion itself (see Chapter 3). More details about the program features can also be found in Schweitzer (1997; 2001; 2002; 2006; 2018; 2025a; 2025b), Schweitzer & Kennett (2007) and Schweitzer *et al.* (2021b).

To invert for a seismic event location, the program reads a set of ASCII-formatted input files. Two files are user defined input files. One file contains the observed data for the event to be localized (**hyposat-in**) and the other file contains a list of parameters to steer this inversion process (**hyposat-parameter**). The location results will be written in the file **hyposat-out**. The contents and structure of these files will be explained in the following three Chapters of this Manual. The program **HYPOMOD** (see for further details Chapter 5) uses the same input files as **HYPOSAT** but writes its results in the file **hypomod-out**.

In addition, there are a number of files containing static input data for the travel-time-table calculations in global spherical Earth models (*e.g.*, **iasp91_A.tbl** and **iasp91_A.hed**), the ellipticity corrections (**elcdir.tbl**), all data defining the global crustal model CRUST 1.0 (**crust1.vp**, **crust1.vs**, **crust1.bnds**) of Laske *et al.* (2013), crustal models of the global models (**std_crusts.dat**), attenuation corrections for magnitude estimations (**MB_G-R.DAT**, **MB_V-C.DAT**, **MB_M-R.DAT**, **MLCORR.TABLE**, **MLCORR.TBALE.wa**, **MLCORR.TABLE.nicolas**), ISC default-depth definitions (**isc_def_depth.dat**, **grn_def_depth.ak135.dat**), the file with the geographic coordinates of the stations **stations.dat** and a list of European seismotectonic units (**REG_L3.DAT**, see *e.g.*, Lindholm *et al.*, 2000). These static files must be either located in the run-directory or in a shared directory defined by the environment variable **HYPOSAT_DATA**. For some files, alternative directories can be set in **hyposat-parameter**; for further details see Chapter 3.

For **HYPOSAT 6.3**, the entire estimation of an initial epicenter was restructured; for further details see Chapter 3.4.2.

The newest version of **HYPOSAT** and **HYPOMOD** (including source code, this Manual, the PDF version of Schweitzer (1997), data files containing travel-time models and station parameters, and several examples) are since **HYPOSAT 6.2** available on GitHub for free download. The software is written in Fortran, C or C++ and was developed and tested with the GNU Compiler

Collection (GCC) on LINUX PCs and Intel® compilers in a WINDOWS 10 and Windows 11 environment. For instructions on installing **HYPOSAT** and **HYPOMOD** on your computer, please consult the documentation on GitHub (<https://github.com/NORSAR-official/Hyposat>).

All older software-package versions, as well as testing the implementation in a SUN (Solaris) environment, are obsolete and are no longer supported. Be also aware, that the previous option to download the software package from NORSAR's anonymous FTP address *ftp.norsar.no* is no longer supported from version **HYPOSAT 6.1 / HYPOMOD 2.1** on.

If executing *hyposat* or *hypomod* with the parameter '**-v**' on the command line (e.g., *./bin/hyposat -v* or *./bin/hypomod -v*) only version number and build date of the program will be written to standard output without any further data processing.

Questions related to program updates and maintenance should be directed to the author.

2 THE *hyposat-in* FILE

HYPOSAT needs as input a file with all the observed data for a specific event: This file has by default the name *hyposat-in*, but another name can be defined in the parameter file *hyposat-parameter*. Be aware that the format of this file has slightly changed in **HYPOSAT 6**. If you have older formatted files, this must be indicated in the *hyposat-parameter* file (see Chapter 2). The data input file must have the following format:

1st line: any title for event identification of maximum 80 characters (used also in the output-file *hyposat-out*).

2nd – to (n+1)th line for n observed onsets. These lines **must** (!) be compatible with the following format (Fortran: ‘a’ is ASCII string, ‘i’ is integer number, ‘f’ is floating-point number, ‘x’ is a separating blank space):

```
(a5,1x,a8,1x,i4,4(1x,i2),1x,f6.3,1x,f5.3,1x,f6.2,3(1x,f5.2),1x,a7,1x,f6.3,1x,f12.2,1x,f7.2,1x,a8,1x,f5.2)
```

station id (a5), phase name (a8), year (i4), month (i2), day (i2), hour (i2), minute (i2), second (f6.3), standard deviation (STD) of the onset time (f5.3), backazimuth (BAZ) (f6.2), standard deviation of the BAZ observation (f5.2), either slowness or ray parameter [s/deg(ree) or s/°] or apparent velocity [km/s] (f5.2) (see the parameter setting with **SLOWNESS [S/DEG]** in the file *hyposat-parameter* in § 3.5.5) and standard deviation of the slowness or the apparent velocity observation in [s/°] or [km/s] (f5.2).

If the type of the P or S onset is unknown, you can use the phase names **P1** or **S1** to indicate that this is the 1st P-type or the 1st S-type onset at this station. Then, the program chooses the right phase name depending on the epicentral distance of the observation and the employed travel-time table.

Following the IASPEI standard, phases named **tx (TX)** , **rx (RX)** , **x (X)** , **Px (PX)** or **Sx (SX)** are assumed to be unknown phase onsets. Then, onset time and slowness will not be included in the inversion. After the inversion, the program will search for the best fitting seismic phase and list it as used phase name.

For each phase name not defined by the applied travel-time model(s), the program searches for the best fitting phase. However, onset time and slowness or ray parameter of such a phase are not used in the inversion, but the BAZ information may be used if the parameter **USE ALL BAZ INFORMATION** is set in *hyposat-parameter* (see § 3.5.4).

If an onset time, a BAZ, or a slowness value is given without its standard deviation, the following default values are used: onset time 2 s; BAZ 30° (40° for LQ and LR); slowness or ray parameter 5 s/° (any apparent velocity in [km/s] input will be internally converted to [s/°]). If BAZ or slowness values are not available, these places can stay blank, or they must be set to ‘-999.’ or ‘-1.’.

Then follows a seven-character long combination of steering flags (a7). These seven steering flags (**1234567**) control the usage of the different onset parameters during the inversion and have the following meanings and options:

- Position **1** the time reading of this onset can be used ('T' or 't') or not used ('_') for the inversion. The travel-time difference between S- and P-type onsets ($t_s - t_p$) can be used for calculating an initial solution for the source time with the Wadati method (Wadati, 1932; 1933) and/or the epicentral distance from the corresponding station. For this, Position **1** must have been set to 'T' or 't' for both onsets.
- Position **2** the BAZ reading of this onset can be used ('A' or 'a') or not used ('_') for the inversion.
- Position **3** the slowness reading of this onset can be used ('S' or 's') or not used ('_') for the inversion.
- Position **4** the time reading of this onset can be used ('D' or 'd') or not used ('_') to calculate travel-time differences as input data for the inversion.
- Position **5** the onset time reading of this onset will be corrected ('R' or 'r') or not corrected ('_') for the crustal structure below a reflection point at the Earth's surface by calculating the travel-time difference for the ray path through the crust in the used global model (as set with **GLOBAL MODEL** in *hyposat-parameter*) and in CRUST 1.0.
- Position **6** the amplitude and period readings of this onset can be used ('M' or 'm') or not used ('_') for magnitude calculations. (Remark: the *hyposat-in* files in older format do not have this flag).
- Position **7** if set to '2', '3', or '4', other global, spherical Earth models (as set with **GLOBAL MODEL 2**, **GLOBAL MODEL 3**, or **GLOBAL MODEL 4** in file *hyposat-parameter*) will be used to calculate the theoretical onset time, the slowness or ray parameter, and their partial derivatives for this onset. With any other character at this place, the standard global Earth model (see **GLOBAL MODEL** in file *hyposat-parameter*) will be used.

If keeping the positions **1–7** blank (' ') the flag combination **TASDRM_** will be automatically used as default value.

Then, one can add the following optional input data: the period of the observed onset (f6.3), the amplitude of the signal in [nm] (f12.2), the signal-to-noise ratio (SNR) (f7.2), an eight-character long arrival-id (a8) and finally the 2nd standard deviation for the onset time reading (see also **DOUBLE SIDED TIME UNCERTAINTY** in *hyposat-parameter*) (f5.2).

The input of the values for amplitude, period, SNR, arrival-id and 2nd standard deviation are optional. The amplitude / period information is only used if both values are larger than 0.0. The SNR is not used in the program but listed in the output files.

All lines starting with 'STOP', a star '*' or a blank space ' ' as well as identical lines following each other are ignored and not used.

When using the correct format, an input file may look like the example below:

----- example of a *hyposat-in* file -----

NORTHERN MOLUCCA SEA, 1996 29 June, converted from pIDC's Reviewed Event Bulletin (REB)

* The row above is the title line

*

```
*23456789 123456789 123456789 123456789 123456789 123456789 123456789 123456789 123456789 123456789 123456789
*aaaa aaaaaaa iiii ii ii ii ff.fff f.fff fff.ff ff.ff ff.ff aaaaaa ff.fff fffffffff ff.fff aaaaaa ff.ff
WRA P 1996 06 29 00 41 44.700 0.300 331.50 20.00 11.10 1.00 T ASD M 0.300 4.50 97.80 abcdef01 0.10
WRA S 1996 06 29 00 45 48.7 0.600 338.0 20.0 17.0 2.00 T ASD M 0.900 2.00 8.10 abcdef02 0.20
QIS P1 1996 06 29 00 42 12.8 0.300 -999. 0.0 -999. 0.00 T D 35.50 abcdef03 0.10
QIS P cP 1996 06 29 00 45 44.8 0.300 T D abcdef04 0.10
ASAR P 1996 06 29 00 42 16.9 0.300 346.3 20.0 7.1 1.00 T ASD M 0.500 3.40 35.10 abcdef05 0.10
ASAR P cP 1996 06 29 00 45 45.2 0.300 345.1 20.0 2.3 1.00 T ASD M 0.500 2.20 11.50 abcdef06 0.10
ASAR S 1996 06 29 00 46 45.3 0.600 347.6 20.0 20.3 2.00 T ASD M 0.800 3.90 6.10 abcdef07 0.20
WARB P 1996 06 29 00 42 31.2 0.300 339.4 30.0 8.2 2.00 T ASD M 0.700 6.50 8.20 abcdef08 0.10
WARB P cP 1996 06 29 00 45 49.4 0.300 -999. 0.0 -999. 0.00 T D abcdef09 0.10
MEEK P 1996 06 29 00 42 42.7 0.300 -999. 0.0 -999. 0.00 T 6.80 abcdef10 0.10
CMAR P 1996 06 29 00 43 09.0 0.300 109.7 30.0 7.8 2.00 T ASD M 0.400 0.60 4.30 abcdef11 0.10
CMAR LR 1996 06 29 00 57 48.7 50.000 110.0 30.0 39.5 2.00 T ASD M 19.360 188.80 abcdef12 25.00
FORT P 1996 06 29 00 43 11.3 0.300 T 10.80 abcdef13 0.10
WOOL P 1996 06 29 00 43 15.6 0.300 7.9 30.0 9.9 2.00 T ASD M 0.600 4.10 8.80 abcdef14 0.10
SHK P 1996 06 29 00 43 25.4 0.300 -999. 0.0 -999. 0.00 T abcdef15 0.10
STKA P 1996 06 29 00 43 46.8 0.300 323.2 20.0 9.0 1.00 T ASD M 0.600 7.20 8.30 abcdef16 0.10
STKA P cP 1996 06 29 00 46 12.0 0.300 -999. 0.0 -999. 0.00 T D abcdef17 0.10
KSAR P 1996 06 29 00 43 46.7 0.300 177.3 20.0 10.1 2.00 T ASD M 0.700 1.70 7.20 abcdef18 0.10
KSAR LR 1996 06 29 01 04 25.0 50.000 160.0 20.0 46.0 2.00 T ASD M 19.860 40.10 abcdef19 25.00
MJAR P 1996 06 29 00 43 51.9 0.300 357.1 20.0 18.8 1.00 T ASD M 0.550 2.00 abcdef20 0.10
PDY P 1996 06 29 00 46 47.0 0.300 111.2 30.0 6.6 2.00 T ASD M 0.450 4.30 23.80 abcdef21 0.10
ZAL P 1996 06 29 00 47 09.0 0.300 -999. 0.0 -999. 0.00 T abcdef22 0.10
ABKT P 1996 06 29 00 48 09.5 0.300 -999. 0.0 -999. 0.00 T abcdef23 0.10
*
```

*NRI P 1996 06 29 00 48 08.7 0.300 195.9 30.0 3.9 2.00 T ASD M 0.750 3.00 11.20 abcdef24 0.10

* Note, the following phase has an unknown phase name! Therefore, its onset time or slowness cannot be used

* for the inversion. However, the backazimuth observation may be usable.

*

```
NRI x 1996 06 29 00 48 08.7 0.300 195.9 30.0 3.9 2.00 T ASD M 0.750 3.00 11.2 abcdef24 0.10
MAW P 1996 06 29 00 49 01.1 0.300 -999. 0.0 -999. 0.00 T abcdef25 0.10
KVAR P 1996 06 29 00 49 17.2 0.300 -999. 0.0 -999. 0.00 T abcdef26 0.10
NPO P 1996 06 29 00 49 31.0 0.300 -999. 0.0 -999. 0.00 T abcdef27 0.10
ARCES P 1996 06 29 00 49 51.7 0.300 94.5 20.0 4.1 1.00 T ASD M 0.550 0.80 10.50 abcdef28 0.10
SPITS P 1996 06 29 00 49 55.9 0.300 116.6 30.0 3.5 2.00 T ASD M 0.900 3.60 10.40 abcdef29 0.10
FINES P 1996 06 29 00 50 00.0 0.300 111.2 20.0 5.9 1.00 T ASD M 0.550 0.60 6.60 abcdef30 0.10
HFS P 1996 06 29 00 50 28.7 0.300 311.3 30.0 1.5 2.00 T ASD M 0.750 1.10 3.70 abcdef31 0.10
TXAR PKPdf 1996 06 29 00 55 42.7 0.300 -999. 0.0 -999. 0.00 T abcdef32 0.10
SCHQ PKPdf 1996 06 29 00 55 41.2 0.300 -999. 0.0 -999. 0.00 T abcdef33 0.10
DBIC PKPdf 1996 06 29 00 55 57.1 0.300 -999. 0.0 -999. 0.00 T abcdef34 0.10
PLCA PKPdf 1996 06 29 00 56 09.1 0.300 311.9 30.0 6.4 2.00 T ASD M 0.900 0.90 4.40 abcdef35 0.10
LPAZ PKPdf 1996 06 29 00 56 44.7 0.300 -999. 0.0 -999. 0.00 T abcdef36 0.10
```

----- end of the *hyposat-in* file example -----

3 THE *hyposat-parameter* FILE

The file *hyposat-parameter* contains the inversion-steering parameters and is read by both programs **HYPOSAT** and **HYPOMOD** (see Chapter 5). The file must be located either in the directory where the program is executed or in the directory defined by the environment variable **HYPOSAT_PARA**. For instructions on how to define environment parameters please consult the program-package-installation documentation on GitHub. If the parameter file has another name than *hyposat-parameter*, this name, including its whole directory path name, must be defined by the environment variable **HYPOSAT_PARA**. This file is read only once at the beginning of a location run.

Several of the parameters are only applicable in special cases. They contribute to the customization possibilities of the program for different applications and the user should check carefully what is really needed for their actual case and what can just be ignored or not set. See the examples in the directory *./examples/* and the file *hyposat-parameter_all* in the directory *./man/* with all possible parameters listed together with their default settings.

HYPOSAT and **HYPOMOD** can read data either from files formatted in their native format or in the International Seismic Format (ISF 1.0) as defined by the Commission of Seismological Observation and Interpretation of IASPEI version ISF 1.0. This format is compatible with the format used by the International Monitoring System (IMS) of the Comprehensive Test-Ban-Treaty Organization (CTBTO) for its bulletins. **HYPOSAT** can also write the inversion results in ISF 1.0 and in JSON formatted output files. For the i/o of ISF formatted files an ISC-software package was adapted for the usage in **HYPOSAT**. Some parameter settings in the *hyposat-parameter* file are only related to i/o of ISF formatted files (see § 3.8).

The syntax in the *hyposat-parameter* file has changed during the development of the different **HYPOSAT** versions. For compatibility reasons, the old syntax versions are still working and will be supported – at least for a while. All parameter settings in old syntax and their new equivalent entries are listed at the end of Chapter 3.

In the following, all parameters and their possible settings are explained in detail. Parameter settings and relevant parts of their definitions are denoted in this document in **bold** and **capital characters**. As reference, the syntax for each parameter setting is given in red with the default parameter setting. For running **HYPOSAT** with specific parameter settings, you can just copy the relevant lines from these examples and modify them as needed. See also the file *hyposat-parameter_all*.

3.1 GENERAL FORMATTING RULES

Each parameter is defined in an extra line, starting with the **mandatory parameter name**, followed by a **required colon** (':') After the **mandatory colon** follows one **mandatory blank space** (' ') and then the parameter setting itself.

Leading blank spaces (' ') in a line as well as all characters between the **parameter name** and the **required colon** will be ignored. *E.g.*, for better readability, a blank space (' ') was added before the **required colon** for all examples of parameter settings in this document.

Be aware that since **HYPOSAT 6.2**, this **mandatory colon** has no longer to be located at the fixed position at character 36 in the line, but it **must** follow after the parameter name. However, all old *hyposat-parameter* files will still work as before.

Any blank line or lines with a star (*) as 1st character are ignored.

Whenever a line does not comply with these rules, a warning message will be written when running the program and the setting in the line will be ignored.

The order in which the parameters are set is arbitrary. Each line can be repeated several times within the file with another setting. In this case, the last set value is applied for the location process. The total number of lines in the file is limited to 2000 lines.

All default values are set when starting the program. For file names, the full directory path name can be given with a maximum length of 512 characters.

3.2 SEISMIC VELOCITY MODELS – TRAVEL-TIME TABLES

3.2.1 Global Models

The parameter **GLOBAL MODEL** (or **GLOBAL MODEL 1**) defines the reference spherical velocity model used to calculate all travel-time related theoretical data. The **HYPOSAT 6.3** package contains data files for the following global models:

ak135_A	AK135 model (Kennett <i>et al.</i> , 1995)
ek137_A	EK137 model (Kennett, 2020)
iasp91_A	IASPEI 1991 model (Kennett, 1991; Kennett & Engdahl, 1991)
iasp91a_A	IASPEI 1991, with a corrected core model (Kennett, pers. communication)
jb_A	Jeffreys-Bullen model (Jeffreys & Bullen, 1940 and later)
prem_A	PREM model (Dziewonski & Anderson, 1981)
sp6_A	SP6 model (Morelli & Dziewonski, 1993)

Some additional, regional spherical Earth models with different crusts and uppermost mantle layers are also included in the **HYPOSAT** software package. From a depth of 220 km on, they are identical to the AK135 model:

barey_A	BAREY, regional model for the European Arctic (Schweitzer & Kennett, 2007)
barez_A	BAREZ, regional model for the European Arctic (Schweitzer & Kennett, 2007)
barents16_A	BARENTS16, regional model for the western Barents Sea (Pirli & Schweitzer, 2017)
bergen_A	University of Bergen's velocity model for Norway (Havskov & Bungum, 1987)
fescan_A	NORSAR's Fennoscandia model (Mykkeltveit & Ringdal, 1981)

The directory, where these data files are located, must be specified with the environment variable **HYPOSAT_DATA** before starting the program. The travel-time tables are calculated with the *libtau*-software package (Buland & Chapman, 1983), and originally distributed as IASPEI91-software, here with the extensions described in the **Introduction**. To become computer architecture and operation system independent, all global travel-time models for the *libtau*-software were recalculated for the **HYPOSAT** package, stored in ASCII format and got the suffix '_A' in their table names. The default global model is 'ak135_A'

GLOBAL MODEL : ak135_A

or alternatively:

GLOBAL MODEL 1 : ak135_A

One can give the name of any other 2nd global model to be used for specific ray paths as indicated in the data input file *hyposat-in*. Up to three additional spherical Earth models can be used for specific travel paths.

GLOBAL MODEL 2 : ak135_A

One can give the name of any other 3rd global model to be used for specific ray paths as indicated in the data input file *hyposat-in*.

GLOBAL MODEL 3 : ak135_A

One can give the name of any other 4th global model to be used for specific ray paths as indicated in the data input file *hyposat-in*.

GLOBAL MODEL 4 : ak135_A

The seismic velocities in the crust must be known for the chosen dominant global model (set with the parameter **GLOBAL MODEL**), so that **HYPOSAT** can apply crustal travel-time corrections. The crustal structures of all prepared *libtau*-type models are collected in a file called *std_crusts.dat* and marked with a unique code. This file must reside in the same directory as defined by the environment variable **HYPOSAT_DATA**. *std_crusts.dat* is included in the directory *./data/*. If you use another global model than AK135 as defined by **GLOBAL MODEL** (or **GLOBAL MODEL 1**), the program will automatically adjust this parameter. However, you can overwrite this setting and force the program to use a specific standard crust by setting the following parameter. The following different codes are possible:

- E1 Jeffreys-Bullen (global standard model)
- E2 PREM (global standard model)
- E3 IASPEI 1991 (global standard model)
- E4 IASPEI 1991a (global standard model, with corrected core)
- E5 SP6 (global standard model)
- E6 AK135 (global standard model)
- E7 FESCAN (regional standard model)
- E8 BAREY (regional standard model)
- E9 BAREZ (regional standard model)
- EA BARENTS16 (regional standard model)
- EB BERGEN (regional standard model)
- EC EK137 (global standard model)

By default, the standard code is 'E6' (for AK135).

GLOBAL CRUSTAL MODEL CODE : E6

If using as input a *hyposat-in* formatted file (see Chapter 2), the usage of alternative spherical Earth models can be specified for specific observations. In the case of using an ISF-formatted input file this is not possible. In this case, one can define an epicentral distance in [°], from which **GLOBAL MODEL 2** should be used to calculate the theoretical onset times. However, be aware that the model usage may differ for some observations from iteration to iteration because of a change in the epicentral distance.

The default value is '999.', which means no usage of a 2nd velocity model in the case of an ISF formatted input file.

ISF_2ND MODEL DISTANCE : 999.

3.2.2 Local or Regional Velocity Models

HYPOSAT can combine global velocity models with an epicentral-distance-dependent **LOCAL OR REGIONAL MODEL**. The local/regional travel-time models are either used around the source and/or to calculate travel-time corrections for reflection points of phases at the Earth's surface or for crustal differences between the dominant global model and the actual structure at the stations (velocities & elevation). The input is a file containing the model parameters. The velocity model must contain the following information:

- In the 1st line maxdis = maximum distance in [°] for which this model shall be used.
- It is followed by lines with depth in [km], the P velocity (Vp) in [km/s], and the S velocity Vs in [km/s]. The model may contain layers with a constant velocity or with velocity gradients.
- First order discontinuities must be specified with two lines for the same depth.
- Additionally, the Conrad and the Moho must be marked as shown in the following example. Please note that the mark for the Conrad and the Moho must be in the 1st line defining the first order discontinuity. Without these discontinuity definitions, the program will name all calculated crustal and regional phases Pg (or Sg, respectively).

CRUST 1.0 defines the Moho but not the Conrad. Here **HYPOSAT** is setting the Conrad at the depth where the P-velocity $V_p \geq 6.3$ [km/s].

All direct phases from sources below a depth of 210 km or with a ray path in a layer below this depth will be named P (or S, respectively) instead of Pn (Sn).

The maximum number of model layers is set to 100.

```

-----example for a file containing a regional velocity-----
10.          | maxdis in free format
   0.000      5.400      3.100      | depth, Vp, Vs in format (3F10.3)
  10.000      5.800      3.200
  20.000      5.800      3.200CONR | + mark for the 'Conrad'
  20.000      6.500      3.600      | in format (3F10.3,A4)
  30.000      6.800      3.900MOHO | + mark for the 'Moho'
  30.000      8.100      4.500      | in format (3F10.3,A4)
   77.500      8.050      4.400
  120.000      8.100      4.500
-----end of example-----

```

If the file containing this velocity model is not available in the run-directory or the directory defined by the environment variable **HYPOSAT_DATA**, one must give the full file name including its directory path. Set it to '_' if no local model is used.

LOCAL OR REGIONAL MODEL : _

Travel times will be calculated for the following seismic phases (as far as the ray theoretically exists with respect to distance and source depth):

- Direct and reflected P-phases: Pg, Pb, Pn, P, pPg, pPb, pPn, pP, PbP (*i.e.*, in this program, the upper side reflection from the Conrad), PmP, PgPg, PbPb, PnPn, PP, PgPgPg, PbPbPb, PnPnPn, PPP
- P-to-S converted phases: pSg, pSb, pSn, pS, PbS, PmS
- Direct and reflected S-phases: Sg, Sb, Sn, S, sSg, sSb, sSn, sS, SbS, SmS, SgSg, SbSb, SnSn, SS, SgSgSg, SbSbSb, SnSnSn, SSS
- S-to-P converted phases: sPg, sPb, sPn, sP, SbP, SmP

When using the local/regional model for calculating local/regional travel-time tables and their partial derivatives, one can define the seismic phase types to be considered by a 4-digit number code: The position of a digit defines the phase type and the value of the digit defines the action for this phase:

dxxx the digit (d) in the 1st position is the flag related to surface reflections (*e.g.*, pP or sS).

xdxx the digit (d) in the 2nd position is the flag for surface multiples (*e.g.*, PP, SS, PPP or SSS).

xxdx the digit (d) in the 3rd position is the flag for reflections at the Conrad or the Moho (*e.g.*, PbP or SmS). Note that here the name 'PbP' is used to indicate a reflection from the Conrad, which is not a regular phase name as recommended by IASPEI (Storchak *et al.*, 2002; 2003; 2006; 2009; 2011; Bormann *et al.*, 2013; Schweitzer *et al.*, 2021a).

xxxd the digit (d) in the 4th position is the flag for converted phases (*e.g.*, sP or PmS).

The digit **d** itself can have the following values:

- 1** = only P-type onsets will be calculated
- 2** = only S-type onsets will be calculated
- 3** = both phase types (P and S) will be calculated

E.g., the combination **1320** means the phases pP, PP, PPP, SS, SSS, SbS and SmS will be calculated, but none of the converted phases. The combination **0000** or simply **0** means only the direct P- and S-onsets, but none of the reflected and/or converted phases will be calculated. For any value below **0**, the program will ignore the local/regional model set by **LOCAL OR REGIONAL MODEL**. The default value is set to '3333'.

PHASE INDEX FOR LOCAL MODEL : 3333

The following setting is steering the usage of the 1° x 1° crustal model CRUST 1.0 of Laske *et al.* (2013) together with the other local/regional velocity models. The usage can be chosen by an integer number:

- <= 0** = no usage of CRUST 1.0 or any other local/regional model as defined with **LOCAL OR REGIONAL MODEL** (this is also the default value).
- 1** = only the local/regional model is used as defined in **LOCAL OR REGIONAL MODEL**. The model is also used for eventually correcting teleseismic travel times of near source reflections (depth phases) (pP, pwP, sS, sP, *etc.*).
- 2** = CRUST 1.0 is only used for calculating station corrections and for travel-time corrections for model differences at the reflection points of the phases (*e.g.*, pP, sS, PnPn, sS, P'P' ...). Reflection point corrections **cannot (!)** be calculated for multiple reflections as *e.g.*, pPP, SSS, *etc.*, and if stations or reflection points are located within the maximum distance from the source as defined in **LOCAL OR REGIONAL MODEL**. Then, it is assumed that the **LOCAL OR REGIONAL MODEL** contains already the correct seismic velocities and no further corrections are needed.
- 3** = CRUST 1.0 is only used to calculate travel times to stations with a maximum distance of 1.5° from the source. **LOCAL OR REGIONAL MODEL** is not used.
- 4** = combines options (2) and (3).
- 5** = **LOCAL OR REGIONAL MODEL** is used for near source surface reflections (depth phases) and CRUST 1.0 is used for all other corrections (station and reflection points).
- 6** = CRUST 1.0 is used to calculate reflection-point corrections only, not for station corrections.

CRUST 1.0 : 0

One can indicate to the program that all the stations are within an epicentral distance of less than about 50 km and that a **LOCAL OR REGIONAL MODEL** is given by setting the parameter **VERY LOCAL GEOMETRY** to '1'. Then, the ray-tracing algorithm is adjusted for a denser ray coverage to calculate the travel times for the different local/regional phases and no ellipticity corrections for the travel times are applied.

VERY LOCAL GEOMETRY : 0

3.2.3 Group Velocities

HYPOSAT can use observations from different types of surface waves, for which the mean group velocities in [km/s] can be set by the following parameters:

RG GROUP-VELOCITY : 2.5

LG GROUP-VELOCITY : 3.5

LQ GROUP-VELOCITY : 4.4

LR GROUP-VELOCITY : 3.95

Infrasound onsets and T-phases can also be included. Onset times of these phases are not well defined but mean group velocities can be used to identify such onsets.

The mean infrasound group velocity (celerity) and the T-phase mean group velocity can be set in [km/s] by:

IS-PHASE GROUP-VEL : 0.33

T-PHASE GROUP-VEL : 1.45

3.2.4 Model Uncertainties

HYPOSAT can account for model uncertainties by defining mean uncertainties of travel-time tables. The default value is taking no model uncertainties ('0').

INCLUDING MODEL UNCERTAINTY : 0

As one implemented option, the uncertainties can be given separately for P- and S-type onsets in [s] but are assumed to be distance independent and then equal for all phases of the same phase type (P or S). These uncertainties will be added to the onset-time uncertainties and thereby influence the weighting of the observations and the final uncertainty of the inversion results. The default values are set to 0. s for both P- and S-type onsets.

MEAN P-WAVE MODEL UNCERTAINTY : 0.

MEAN S-WAVE MODEL UNCERTAINTY : 0.

Another option is to define uncertainties in [s] for some specific phases separately. For now, this is possible for the following regional phases: Pg, Pb, Pn, Sg, Sb, Sn and Lg. The default values are again set to 0. s.

MEAN PG TRAVEL-TIME UNCERTAINTY : 0.

MEAN PB TRAVEL-TIME UNCERTAINTY : 0.

MEAN PN TRAVEL-TIME UNCERTAINTY : 0.

MEAN SG TRAVEL-TIME UNCERTAINTY : 0.

MEAN SB TRAVEL-TIME UNCERTAINTY : 0.

MEAN SN TRAVEL-TIME UNCERTAINTY : 0.

MEAN LG TRAVEL-TIME UNCERTAINTY : 0.

3.3 STATION RELATED PARAMETERS

The program needs one file with the geographic coordinates of the sensor locations (employed station coordinates). Four file formats are supported: PDE (NEIC, see also the file ***data/stations.dat*** in the program package), CSS3.0, an ISC-type listing or a comma separated listing. These file formats are automatically detected. To obtain the location results faster, it is recommended to use a file containing only your network stations. The file must be either located in the run-directory, or the directory defined by the environment variable **HYPOSAT_DATA**, or one has to give the full file name including directory path information. The default file name is '***stations.dat***'.

The four possible format definitions (Fortran) are, with examples:

PDE (NEIC) format:

station id, flag, lat [degrees], lat [minutes], lat [seconds], N/S, lon [degrees], lon [minutes], lon [seconds], E/W, elevation [m], name
(a5,a1,i2,i2,f4.1,a1,i3,i2,f4.1,a1,f7.1,a48)

```
NOA    610223.0N 111253.1E  717.0NORSAR Array
```

CSS3.0 format:

station id, start time, end time, lat [°], lon [°], elevation [km], name
(a5,2x,i8,1x,i8,1x,f9.4,1x,f9.4,1x,a48)

```
NOA      1976275  9999365   61.0397   11.2148   0.7170 NORSAR Array
```

ISC-type listing format:

1st title line
2nd title line
station id, lat [°], lon [°], elevation [m]
(a5,1x,3(1x,f10.0))

```
NOA      61.0397   11.2148   0.7170
```

Comma separated ISC-type listing:

station id, station id, lat [°], lon [°], elevation [m]

```
NOA, NOA, 61.0397, 11.2148, 0.7170
```

STATION FILE : ./stations.dat

In the case that one has two different station-coordinate lists, one can define an alternative station file, which the program will check for station coordinates if it cannot find the station in the file defined by the parameter **STATION FILE**. With this option one can also combine the usage of station lists if they have two different file formats. The default file name is also '**stations.dat**'.

ALTERNATIVE STATION FILE : ./stations.dat

There are several possibilities to correct onset times for station specific effects. For details on how the estimation of such corrections is implemented in **HYPOSAT** see Schweitzer (2025). This parameter is a flag on whether station corrections (elevation, crust) will be applied. By default, station corrections are taken in account '1'.

STATION CORRECTIONS : 1

One can set global crustal velocities in [km/s] for estimating elevation corrections with these parameters. If the parameter **S-VELOCITY TO CORRECT ELEVATION** is not set or set to 0. and the parameter **P-VELOCITY TO CORRECT ELEVATION** is defined earlier in the file, the **S-VELOCITY TO CORRECT ELEVATION** is calculated from the **P-VELOCITY TO CORRECT ELEVATION** value divided by $\sqrt{3}$. All values below 0.001 and above 99. km/s are ignored and the AK135 values are used as default.

E.g., for AK135:

P-VELOCITY TO CORRECT ELEVATION : 99.1

S-VELOCITY TO CORRECT ELEVATION : 99.1

or

P-VELOCITY TO CORRECT ELEVATION : 5.80

S-VELOCITY TO CORRECT ELEVATION : 3.46

E.g., for Jeffreys-Bullen:

P-VELOCITY TO CORRECT ELEVATION : 5.56

S-VELOCITY TO CORRECT ELEVATION : 3.37

Station specific onset time corrections can be added in an extra file. If local seismic velocities at the stations are known, one can list them in this file and then, these values will be used to calculate station-elevation effects. Also fixed station corrections can be additionally used to correct for constant time shifts and are listed in this file.

These corrections are separate for P and S waves. For more details on station-correction calculations see Schweitzer (2025). The velocity value can also be used to correct for a known velocity anomaly below a station. The input is format free. If a station is not on this list, the default velocity values as defined by the input parameters **P-VELOCITY TO CORRECT ELEVATION** and **S-VELOCITY TO CORRECT ELEVATION** are used. In the case that one wants to use only static time corrections (*e.g.*, because of a known digitizer problem) but no special velocity values for elevation corrections, one can set the velocities to **0.** and only add the static time corrections.

```
----- example for an ASCII file containing station corrections -----
* This is a file containing values to calculate station corrections (only
* for the stations listed here!).
*
* All lines starting with a star '*' or empty lines are ignored.
*
* Minimum input:
* Station name and local P and S velocities to calculate elevation effects
* Vp or Vs given in this file for a specific station always overrides values
* given by the parameters P-VELOCITY TO CORRECT ELEVATION and
* S-VELOCITY TO CORRECT ELEVATION or defined by CRUST 1.0:
*
* Stat Vp Vs
GERES 5.2 3.2
*
* Optional input (e.g., to correct for results from tomographic studies
* or known clock errors) of static time shifts. Static corrections for P-
* and S-body waves.
* No corrections are applied to IS- or T-phase onsets. These correction
* values will be added (!) to the observed travel times:
*
* Stat Vp Vs dtp dts
GERES 5.2 3.2 0. 0.
*
* Additional static time corrections for surface-wave type onsets (LQ, LR, Lg, Rg).
* The corrections are added (!) to the observed travel times:
*
* Stat Vp Vs dtp dts dtsurface
GERES 5.2 3.2 0. 0. 0.
* In the case that one wants to use only static time corrections, one can
* set the velocities to 0. (then they will be ignored) and only the static
* time corrections are used:
*
* Stat Vp Vs dtp dts
XYZ 0. 0. 1.0 1.0 1.0
----- end of example-----
```

CAUTION: the constant time shifts, as defined in the correction file, are used to calculate travel-time residuals, but they are not changing (correcting) the observed onset times as given in *hyposat-in* or written in the output files with the **HYPOSAT** location results.

STATION CORRECTION FILE : _

By default, static station corrections are used for all onsets from the particular station. If the static station corrections in the **STATION CORRECTION FILE** are only valid for the 1st onsets of P or S phases (if *e.g.*, calculated with *VELEST* (see Kissling, 1988; 1995), this switch must be set to '1'.

STATION CORR ONLY 1ST PHASE : 0

In the case of a very local geometry, stations may be located below the surface (*e.g.*, in boreholes or mines) or even below the event. Then, this has to be taken in account when calculating the theoretical travel times and one must set this switch to '1'. In this case the parameter **VERY LOCAL GEOMETRY** is also set automatically to '1'. By default, this option is switched off ('0').

LOCAL STATION BELOW SURFACE : 0

One can consider defining a maximum and a minimum epicentral distance of stations to be used during the inversion. Both distances must be given in [°]. If the inversion output is set with **OUTPUT IN KM** to [km] (see § 3.8.3), both **MIN EPI DIST OF STAT** and **MAX EPI DIST OF STAT** are internally converted to [km].

MIN EPI DIST OF STAT : 0.

MAX EPI DIST OF STAT : 180.

3.4 SOURCE RELATED PARAMETERS

The program has different ways to apply or calculate a preliminary source time and a preliminary epicenter. This can be steered with different parameters. In any case, any initial value directly given in the parameter list, will overwrite all estimated values. The following parameters define the initial solution including its uncertainties:

3.4.1 Initial Source Time

To set an initial source time the accepted formats are:

Epochal time	(which means number of seconds since 01/01/1970 00:00:00.0) allowed is any number larger than -2840140800.0 (= default source = 1 January 1880 00:00:00.0). Any smaller number will be automatically set to -2840140801.0.
yyyy-doy?hh?mi?ss.sss	with doy (day of the year), and for '?' one can use any character including a blank space ' '. The seconds can be omitted.
yyyy-mm-dd?hh?mi?ss.sss	The seconds can be omitted.

Any time earlier than the default value will be ignored. Here are some examples for the same time in the different allowed formats:

STARTING SOURCE TIME : 1970-001:00.00.00.0

STARTING SOURCE TIME : 1970-001 00 00 00.0

STARTING SOURCE TIME : 1970-01-01 00:00:00.0

STARTING SOURCE TIME : 1970-01-01 00 00 00.0

STARTING SOURCE TIME : 0.

Together with an initial source time one can set the source time uncertainty in [s].

STARTING TIME UNCERTAINTY : 10.

If no **STARTING SOURCE TIME** is set, an initial source time will be estimated from the travel-time differences between direct S-type and direct P-type observations ($t_s - t_p$) by using Wadati's approach (Wadati, 1932; 1933). For this, the program calculates mean V_p/V_s ratios for each observed phase type and estimates a source time, respectively. The phase types (type 1: P & S, type 2: Pg & Sg, type 3: Pn & Sn and type 4: Pb & Sb) are identified based on the phase names of the data input. For cases including the phase names P1 or S1 one of the 4 types is chosen depending on the travel-time differences in the global model AK135. Then, the initial source time is calculated as the mean value of all estimated source times.

In the case of only one $t_s - t_p$ observation, the V_p/V_s ratio is set to $\sqrt{3}$ with an uncertainty of 0.2.

If no $t_s - t_p$ observation is available and no source time set (see above) the starting source time is set to the earliest observed onset time.

The Wadati approach can be steered by the length of the allowed $t_{s\text{-type}} - t_{p\text{-type}}$ travel-time differences measured in [s]. By default, the minimum allowed travel-time difference is set to 0 and the maximum to 300 s.

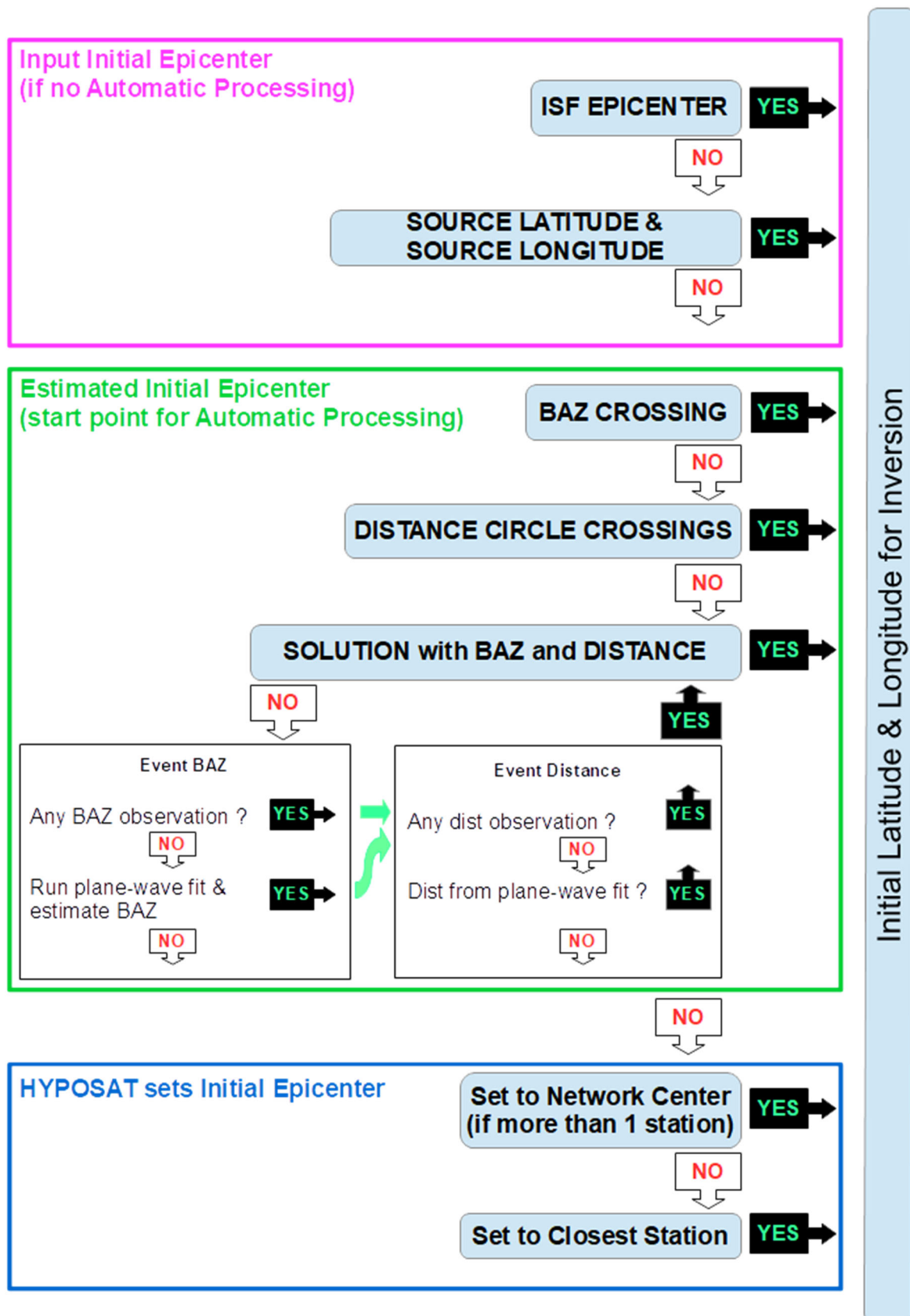
MIN DT FOR WADATI : 0.

MAX DT FOR WADATI : 300.

3.4.2 Initial Epicenter Estimation

For each location algorithm, the chosen initial epicenter can have strong influence on the location results. **HYPOSAT** has different options to insert an initial epicenter that follow a specific hierarchy. An initial epicenter can be set by input or estimated based on the input data. In the unlikely case that no initial solution is set or can be estimated, the initial epicenter is set to the center of the station network.

The flow chart shows the implemented hierarchy for the case of event locations with a network of single stations and/or arrays. The different options for initial epicenter estimation are described in more details below and can be steered by parameters.



Flow chart of the cascade of different options to define an initial epicenter for the inversion.

The geographic coordinates of an initial epicenter can be directly set, together with their uncertainties. This setting will overwrite (unless the parameter **AUTOMATIC PROCESSING** is set, see below) evaluations based on crossing BAZ observations, the results of a plane wave approximation, or a single array solution.

Latitude

The allowed range is from -90 to +90 degrees, all other values as well as a ‘_’ are ignored and then no starting source latitude is set.

STARTING SOURCE LATITUDE : 999.

or

STARTING SOURCE LATITUDE : _

and

STARTING LATITUDE UNCERTAINTY : 10.

Longitude

The allowed range is from -180 to +180 degrees, all other values as well as a ‘_’ are ignored and then no starting source longitude is set.

STARTING SOURCE LONGITUDE : 999.

or

STARTING SOURCE LONGITUDE : _

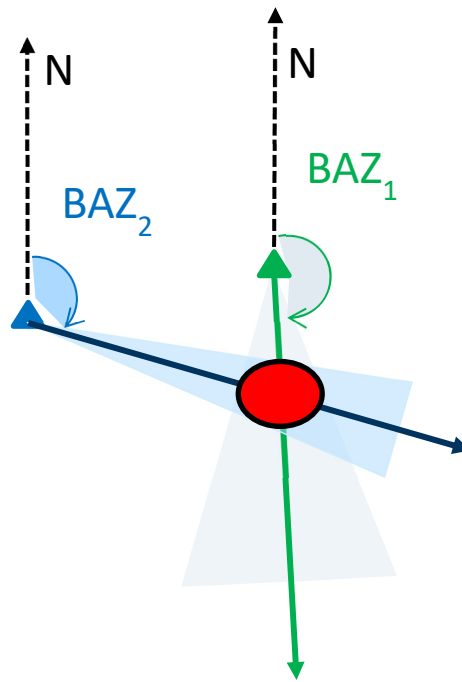
and

STARTING LONGITUDE UNCERTAINTY : 10.

In the case of an ISF formatted input file, the input file already contains source location and source time. This epicenter can be used as initial (and reference) solution. By default, the ISF epicenter is ignored and the flag is switched off, ‘0’.

ISF EPICENTER : 0

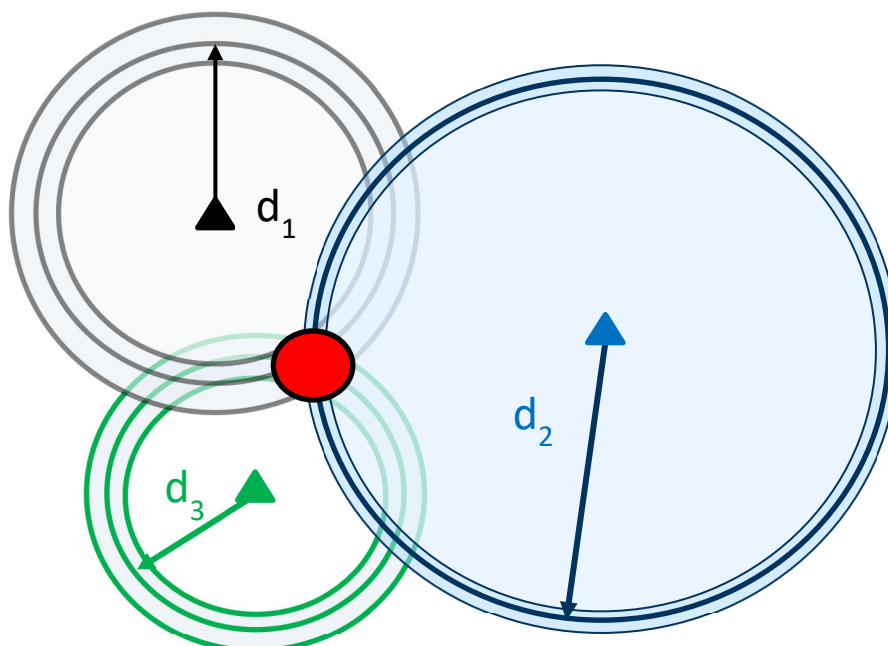
If BAZ observations are available from several stations or arrays, these data (*i.e.*, mean value of all BAZ observations per station) are used to calculate an epicenter by calculating all possible crossing points of these different observations.



An initial epicenter including its uncertainties based on the observation uncertainties is then calculated as the mean value of all crossing points and their spread. By default, the program tries to use this algorithm and the flag to override the option is switched off, '0'.

NO BAZ CROSSING : 0

The travel-time difference between P-type and S-type onsets at the same station is a measure of the epicentral distance. With observations from at least 3 different stations, one can estimate an initial epicenter from the crossing point of the corresponding distance circles around these stations.



Taking the uncertainty of the estimated epicentral distances and hypocentral depths in account, one can estimate possible source regions and their sizes. Since **HYPOSAT 6.3**, this option is implemented as additional algorithm to estimate an initial epicenter. With many such distance-circle observations available, the calculation of all possible crossing points may take quite long. The analysis starts with the stations listed at the beginning of the data input file *hyposat-in* and stops latest when more than 5 distinct source regions were found. Then, the crossing point region with the maximum number of observations is chosen as initial epicenter. By default, the program tries to use this algorithm if no BAZ crossing points are found and the flag overriding its use is switched off, '0'.

NO DIST CIRCLE CROSS : 0

In the case of a single array with BAZ observations and P- and S-onset observations, the program uses these data to estimate a single array location (see *e.g.*, page 37 in Schweitzer *et al.*, 2012) based on the epicentral distance from the travel-time difference between the phases or the observed slowness or ray parameter. By default, the program tries this option as third step and the flag overriding its use is switched off, '0'.

NO REG ARRAY START SOLUTION : 0

If none of the above algorithms give an initial epicenter but P-type onset observations from several stations are available, the program can try a plane-wave fit through the 1st P-wave onsets at all stations to get a BAZ direction to the epicenter. Then, if possible, S-P travel-time differences or the inversion of the slowness or the ray parameter are used to estimate an epicentral distance and a start epicenter is calculated. By default, this option is switched off '0'.

NO PLANE WAVE APPROX : 0

The following parameter brings the program in an automatic processing mode. In this case, the program does not use, in a first step, any initial solution directly set by external input, but accepts initial solutions determined through the hierarchy in the green frame of the flow chart above. However, if the program cannot determine any initial solution, potentially available ones are used as backup. If this switch is set, all onsets in an ISF formatted output file will be marked with an 'a' as automatically measured. By default, automatic processing is switched off, '0'.

AUTOMATIC PROCESSING : 0

3.4.3 Initial Source Depth

One can set an initial source depth in [km] and its uncertainty, respectively. The default depth in the program is 0 km with an uncertainty of 20 km.

STARTING SOURCE DEPTH : 0.

and

STARTING DEPTH UNCERTAINTY : 20.

In the case of an ISF formatted input file the depth solution can be used as start (and reference) solution. By default, this flag is switched off, '0'.

ISF DEPTH : 0

An allowed depth range in [km] can be set. Be aware that the program does not allow any sources with negative depth, which may happen *e.g.*, in models with topography above the mean sea level.

MINIMUM DEPTH : 0.

MAXIMUM DEPTH : 800.

The depth of the event can stay fixed, or the program can try to determine the depth based on the observed data and the location of the epicenter. From **HYPOSAT 6.1** onwards, the program can also use the International Seismological Centre (ISC) default-depth definitions (see Bondár & Storchak, 2011). This can be steered with the parameter **DEPTH FLAG**, which takes the following values

- f (or F)** = depth is fixed at **STARTING SOURCE DEPTH**.
- d (or D)** = a new depth is determined from the beginning for each iteration step.
- b (or B)** = both, the inversion starts with the fixed depth and when an epicenter is found, the inversion continues with inverting also for the depth.
- i (or I)** = the ISC default depth (Bondár & Storchak, 2011) for this epicenter is used as fixed depth.
- j (or J)** = as **b (or B)**, but the starting depth is the ISC default depth.

By default, this parameter is set to 'F'.

DEPTH FLAG : F

An additional steering parameter is needed for choosing the default depth if the parameter **DEPTH FLAG** is set to i, l, j or J. Wherever no default ISC-depth information is available, the focal depth can be fixed relative to the Moho depth as defined in CRUST 1.0. There are 3 different default depth-data sets available, which can be differently combined:

- i) default depth from ISC well-located seismicity, in a geographical grid with 0.5° x 0.5° resolution (*isc_def_depth.dat*)
- ii) default depth derived from the crustal thickness (Moho depth in CRUST 1.0)
- iii) default depth as defined by ISC for the different Flinn-Engdahl Regions (*grn_def_depth.ak135.dat*)

Only options (ii) and (iii) cover the entire globe. Therefore, one has to define how to choose the default depth type with another parameter:

- 1** = search in (i) and if not successful, continue with (ii)
- 2** = search in (i) and if not successful, continue with (iii)
- 3** = search only in (ii)
- 4** = search only in (iii)

By default, this parameter is set to '1'.

DEFAULT DEPTH TYPE : 1

The following parameter defines how the default depth is extracted from CRUST 1.0 (see **DEFAULT DEPTH TYPE**). Here we use only the solid part of the crust (removing any water layer on top) without topography for calculating the depth and afterwards (if existing) adding the thickness of the water layer again, to avoid a source in the water layer.

- uc** = upper crust; 0.25 * CRUST 1.0 Moho depth
- mc** = middle crust; 0.50 * CRUST 1.0 Moho depth
- lc** = lower crust; 0.75 * CRUST 1.0 Moho depth
- mo** = Moho depth; CRUST 1.0 Moho depth

By default, this parameter is set to 'mc'.

CRUST 1.0 DEPTH TYPE : mc

Based on model CRUST 1.0, one can force the program to not accept any source depth in a water layer at the surface (*e.g.*, ocean, big lake). Then, the following flag has to be set to '1' (default value is '0', *i.e.*, no check for a water layer on the top of the model).

CHECK DEPTH FOR WATER LAYER : 0

3.5 STEERING THE USAGE OF OBSERVATIONS

3.5.1 Onset Time Observations and Phase Names

By default, **HYPOSAT** uses travel-time differences between different onsets at the same station as additional constraint. Setting this flag to '0' will overwrite the flag setting 'D' or 'd' for all observations in the input file *hyposat-in* and switch off the usage of travel-time differences during the inversion.

FLAG USING TRAVEL-TIME DIFFERENCES : 1

In the case that travel-time branches of different phases overlap, the correct phase identification can be problematic. To avoid the effect of wrong interpretations, the allowed theoretical travel-time difference between phases of the same phase type is limited to a minimum of 2.0 s. This value can be changed:

MIN ALLOWED TRAVEL-TIME DIFFERENCE : 2.0

As widely known, the reading of seismic onset times do not follow a Gaussian distribution. **HYPOSAT** can use two different standard deviations (STD) for the onset time to weight travel-time-data residuals, depending on the sign of the residual (always observation - calculation). The 1st STD will then be used for positive (onset too late with respect to the theoretical value) and the 2nd STD for negative residuals (onset too early with respect to the theoretical value). For more details see Schweitzer (2006). By default, this option is switched off, '0'.

- 0** = no 2nd STD, the program uses only one STD (see the *hyposat-in* description)
- 1** = reading of a 2nd STD from input file (see the *hyposat-in* description)
- 2** = use the **DBLE SID. UNCERTAINTY FACTOR**

DOUBLE SIDED TIME UNCERTAINTY : 0

If the flag **DOUBLE SIDED TIME UNCERTAINTY** is set to '2', the 2nd STD is defined as (STD * **DBLE SID. UNCERTAINTY FACTOR**). The default value of this factor is '1.'.

DBLE SID. UNCERTAINTY FACTOR : 1.0

One can change the weight of whole groups of observations. This can be steered by a 4-digit code:

- 1st digit: travel-time data
- 2nd digit: BAZ observations
- 3rd digit: slowness observations
- 4th digit: travel-time-difference observations

The digits can have the following values:

- 0** = no change of input uncertainties (the weighting only depends on the given uncertainties).
- 1** = the data uncertainty for the inversion is (dynamically) increased by the residual of this observation (observation is getting a lower weight).
- 2** = the data uncertainty is reduced by a factor of 0.5 (observation is getting a higher weight).
- 3** = the data uncertainty is increased by a factor of 2 (observation is getting a lower weight).

E.g., 0203 would give all BAZ observation higher weight and reduce the weight of all travel-time-difference observations. The default value is '0000' for all digits, which means no weighting change.

CHANGE WEIGHTING TASD : 0000

If an event location should be only based on P-type onsets, one sets this here. By default, this is switched off, '0'.

P-TYPE ONSETS ONLY : 0

One can suppress the usage of Lg or Rg phases as defining observations by setting the following parameters to '0'. By default, they are switched on '1' (if present, Lg and Rg are used).

REGIONAL SURFACE WAVES LG : 1

REGIONAL SURFACE WAVES RG : 1

The usage of the phase names Sg and Lg is not always consistent. Following the IASPEI recommendations for seismic phase names (Storchak *et al.*, 2002; 2003; 2006; 2009; 2011; Bormann *et al.*, 2013; Schweitzer *et al.*, 2021a), **HYPOSAT** assumes Lg to be a surface wave with a given group velocity (see **LG GROUP-VELOCITY**).

The following parameter settings can be used to homogenize the input by automatically changing the phase names for all Sg or Lg observations.

By default, both options are switched off, '0'. If both flags are set to '1', they will be both ignored.

LG-PHASE TO SG : 0

SG-PHASE TO LG : 0

If one wants to use both the travel times of Sg-type onsets as body-waves and the Lg onsets as surface waves, it can make sense to separate these two phases by the epicentral distance. If the following distance in [km] has a value above zero, this epicentral distance is used as separation between Sg and Lg. Then, all Lg are renamed to Sg if the epicentral distance is shorter and all Sg are renamed to Lg if the epicentral distance is equal or larger than this distance.

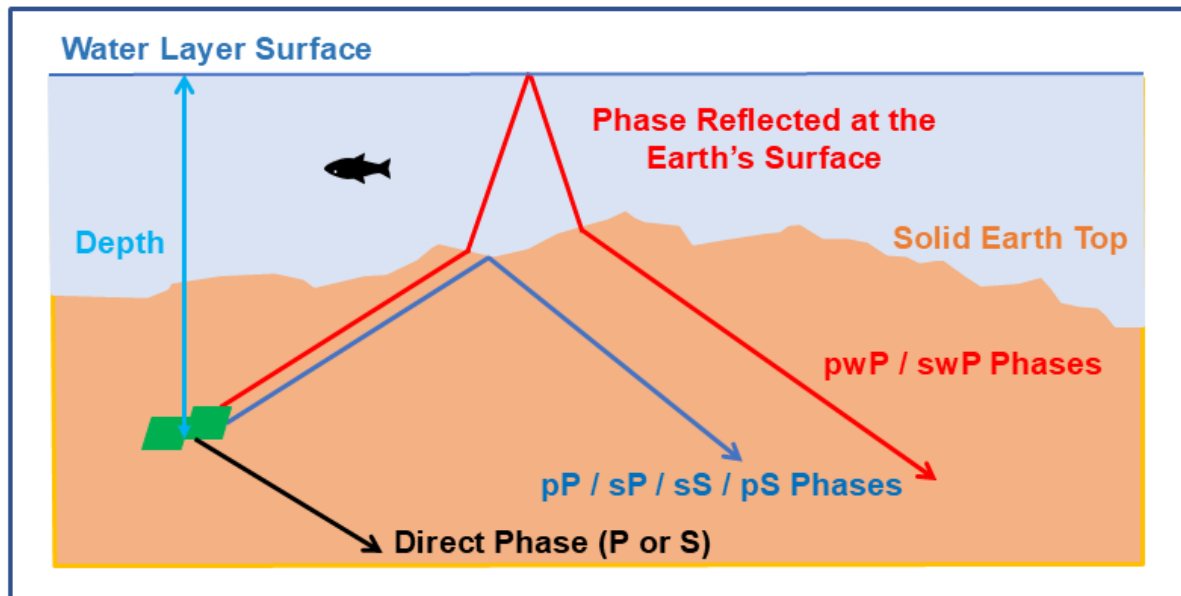
SG--LG DISTANCE : -999.

If the onset times of long-period surface waves should be used as defining observations, one must set this switch to '1'.

LP SURFACE WAVES (LQ/LR) : 0

3.5.2 Surface Reflections in the Case of a Water Layer on Top of the Model

In the case of a **WATER LAYER ON TOP** of the model, **HYPOSAT** can correct the travel times (see also parameter settings for CRUST 1.0 and detailed explanations in Schweitzer (2025)) for the water layer and eventually add a 'w' to the phase name. The water-layer phase is only added for P-type phases and not for any type of converted phases (*e.g.*, sP or pS). All S-type reflections and S-to-P or P-to-S converted phases are assumed to be reflected from the solid-fluid interface. The steering parameter can have these values (by default the parameter is set to '0'):



- 0 = the water layer is ignored. This is also the default case. However, the program may correct for the local velocity structure (*e.g.*, topography) as defined with settings for the parameter CRUST 1.0.
- 1 = all (!) reflections at the Earth's surface are corrected for the water layer, *i.e.*, not only the depth phase pP becomes pwP (Hong & Fujita, 1981; Pearce, 1981; Engdahl & Billington, 1986; Schweitzer, 2025), but also *e.g.*, PP becomes PwP. S-type onsets are then reflected at the solid-fluid interface.
- 2 = **HYPOSAT** chooses the phase identification (reflection from the solid-water interface or from the water-atmosphere interface), which minimizes the travel-time residual for P-type reflections; for S onsets as (1).
- 3 = for S-onsets as (1), optimizing for P onsets only if the travel-time difference between the two reflections (solid-water or water-atmosphere interface) is larger than **WATER LAYER DT** seconds. Otherwise, all reflections are assumed to be from the solid-water interface.
- 4 = as (3), but if the travel-time difference is smaller than **WATER LAYER DT** seconds, all reflections are assumed to be from the water-atmosphere interface.
- 5 = the program uses the phase name(s) of the input file without any optimization. However, the program will automatically ignore all 'w' onset paths without a water layer at the reflection point in CRUST 1.0.

WATER LAYER ON TOP : 0

Minimum travel-time difference in [s] to optimize the phase identification in the case of a water layer on top of the model (for details see parameter **WATER LAYER ON TOP**). The default value is 2 s, which approximately corresponds with a water depth of about 1.5 km (see Engdahl *et al.*, 1998).

WATER LAYER DT : 2.

3.5.3 Other Non-traditional Phase Onsets

If infrasound observations (phase name IS) shall be used as defining, this switch must be set to '1'.

IS-PHASE USAGE : 0

If T-phase observations (phase name T) shall be used as defining, this switch must be set to '1'.

T-PHASE USAGE : 0

3.5.4 Backazimuth (BAZ) Observations

The backazimuth (BAZ; azimuth from station to epicenter measured from North clockwise over East) is an event location relevant datum. To make this clearer, we consequently use the term BAZ from **HYPOSAT 6.1** on. If available, BAZ information is by default used as defining observation. This can be switched off by setting this parameter to '0'.

BAZ AS DEFINING : 1

If one wants to use the BAZ observations only for the estimation of an initial solution (from crossing BAZ observations) and not for the further inversion process, one can set the following parameter to '1'. This option is by default switched off, '0'.

BAZ ONLY INIT SOL : 0

If one has many BAZ observations and wants to get a location based only on these observations (e.g., several infrasound array observations) one can set this flag. Then, **HYPOSAT** calculates a mean epicenter solution and its standard deviations from all possible crossing points of the BAZ observations. No travel-time information is used. This option is by default switched off, '0'.

BAZ ONLY LOC : 0

The next parameter defines the maximum allowed travel-time residual in [s] of a body wave and short-period, regional surface wave (Lg, Rg) onset, for which the BAZ is used as defining observation. This setting is ignored if **USE ALL BAZ INFORMATION** is set to '1'. The default value is 30. s.

MAX T RES FOR BAZ OF B USE : 30.

The next parameter defines the maximum allowed travel-time residual in [s] of a long-period surface-wave onset, for which the BAZ is used as defining observation. This setting is ignored if **USE ALL BAZ INFORMATION** is set to '1'. The default value is 180. s.

MAX T RES FOR BAZ OF L USE : 180.

If setting the following switch to '1', the observed BAZ of all phases (also the not identified but associated ones as listed in *hyposat-in*) will be used as defining during the inversion. This option overwrites **MAX T RES FOR BAZ OF B USE** and **MAX T RES FOR BAZ OF L USE**. However, setting the option **BAZ ONLY INIT SOL** to '1' overwrites this flag. By default, this is switched off, '0'.

USE ALL BAZ INFORMATION : 0

3.5.5 Slowness Observations

The slowness or ray parameter of an observed onset is either defined by its inverse, the apparent velocity in [km/s] or as spherical slowness or ray parameter in [s/°]. The program handles both, but in one input file (see *hyposat-in*), only one unit type can be used. By default, and if the input file is in ISF format, the unit is set to [s/°]. If the apparent velocity is provided in *hyposat-in* this parameter must be set to '0'.

SLOWNESS [S/DEG] : 1

If the slowness observations should not be used as defining, the next switch has to be set to '0'. By default, the program uses the slowness observations '1'.

SLOWNESS AS DEFINING : 1

The following parameter defines the maximum allowed travel-time residual for which the slowness is used as defining observation for body waves. The default value is set to 10 s.

MAX T RES FOR SLOWNESS USE : 10.0

3.5.6 Other Settings for Input Data

If a seismic phase of an onset has a large residual with respect to the theoretical onset time and another seismic phase has a smaller residual, the program may try a new solution after (internally) changing the phase name. The program can also automatically declare observations as undefining (see also the parameter settings for, **MAX T RES FOR BAZ OF B USE, MAX T RES FOR BAZ OF L USE, MAX T RES FOR SLOWNESS USE**). However, this behaviour is not always desired by the user. If the following flag is set, all observations are used for the inversion according to the setting of **TASDR** (see detailed description in *hyposat-in*). In this case, phase names remain unchanged and used as provided in *hyposat-in*, serving as defining observations regardless of the size of the travel-time residuals. BAZ and slowness or ray parameter observations are used as defining, as long as they have residuals less than 50° and 10 [s/°], respectively.

If this flag is set, **CONSTRAIN SOLUTION** (see § 3.6) is automatically switched off, '0'.

By default, this option is switched off, '0'. However, in the case of less than 4 observed phases, this flag is automatically set to '1'. This parameter setting is not overwriting the search for the best fitting phase name in the case of using P1 or S1 as input phase names.

FLAG USING INPUT FIXED : 0

The program can check the input for more than one onset of the same phase at a given station and reduce these onsets to one entry. This option is of particular interest whenever onset readings from several sources are merged to one single input file. The reduction (after removing double entries) is done by calculating mean onset parameters and adding the spread of the different onset parameters (time, BAZ and slowness) to the reading uncertainty of this observation.

Any amplitude / period readings will not be changed. By default, this option is switched off, '0'.

FLAG CHECKING MULTIPLE ONSETS : 0

If **FLAG CHECKING MULTIPLE ONSETS** is set, the maximum time difference between two observations to be merged into one new observation in [s] has also to be defined.

MAX DT FOR MULTIPLE ONSETS : 5.0

3.6 THE INVERSION PROCESS STEERING PARAMETERS

The maximum number of iterations is arbitrarily set by default to 80 iterations.

MAXIMUM # OF ITERATIONS : 80

The iterations stop when two consecutive solutions are separated by less than the given accuracy in [km].

LOCATION ACCURACY : 1.0

In some cases, the iteration results oscillate between two or more solutions. The following parameter defines how many iterations back we search for such oscillations. The maximum number allowed is 15; the default value is 4. In the case of an oscillating solution, the program output will internally try to jump out of this loop by changing uncertainties, data used as defining or fixing the depth. If this does not result in a stable solution, the program will list the parameter range of solutions.

ITERATIONS TO SEARCH OSCILLATIONS : 4

In the case that beside onset times also BAZ and slowness or ray parameter observations are used to invert for the source location, the mean residual of all defining travel times can become unequal 0. With this parameter, the program is forced to adjust the source time for this mean shift. By default, this option is switched on '1'.

MEAN T-T RES. CORREC : 1

The last iteration can constrain the solution by concentrating on the most trustable/important data and removing the other from the inversion process by declaring them as undefined (see data input in *hyposat-in*). Five different cases are implemented. If **FLAG USING INPUT FIXED** is set to '1', this parameter setting is ignored. By default, this option is switched off ('0').

- 0** = no final constraining
 - 1** = final constraining of the solution by removing data with large residuals
 - 2** = final constraining by removing unimportant data as defined by the Information Density Matrix (see the Generalized Matrix Inversion (GMI) theory *e.g.*, in the textbooks of Aki & Richards (1980) or Menke (1989))
-

- 3 = case 1 & 2
- 4 = case 1 with fixed max residual (default 30 s)
- 5 = case 4 & 2

CONSTRAIN SOLUTION : 0

An additional parameter is needed for the cases 2, 3, 5 of **CONSTRAIN SOLUTION**, to define the threshold for low importance input data. All data with Information Density Matrix diagonal-element entries smaller than this threshold are then not used in the inversion. The threshold is calculated as the ratio between the actual Information Density Matrix entry for this datum and the datum with the maximum Information-Density-Matrix value. The default value is 0.005 = 0.5 % of the most important datum.

However, if this value is set to any negative value, the program calculates the threshold to (1 / number of defining data).

INF. DENSITY MATRIX THRESHOLD : 0.005

For the cases 4 and 5 of **CONSTRAIN SOLUTION** additional parameters are also needed. Then maximum residuals for travel-time, BAZ and slowness or ray parameter must be defined. The default values for travel-time residuals are 30 s, BAZ 50° and ray parameter 10 s/°.

MAXIMUM ALLOWED P RESIDUAL : 30.0**MAXIMUM ALLOWED S RESIDUAL : 30.0****MAXIMUM BAZ RESIDUAL : 50.0****MAXIMUM SLOWNESS RESIDUAL : 10.0**

The confidence level of modelled uncertainties is measured in [%] probability. The default value is 68.3 % (*i.e.*, +/- 1 standard deviation).

CONFIDENCE LEVEL : 68.3

3.7 MAGNITUDE CALCULATION PARAMETERS

If the data input file contains amplitude / period readings, station and network magnitudes can be estimated. By default, this option is switched off, '0'.

MAGNITUDE CALCULATION : 0

Three body wave attenuation models are available to calculate mb from 1st P-type onsets. The maximum allowed mb value is 8.5. The default model is set to 'G-R':

- G-R** = Gutenberg-Richter (1956a; 1956b), file **MB_G-R.DAT**
- V-C** = Veith-Clawson (1972), file **MB_V-C.DAT**
- M-R** = Rezapour (2003), file **MB_M-R.DAT**

P-ATTENUATION MODEL : G-R

One can choose between two surface wave attenuation models to calculate MS from LR observations with a signal period larger than 5 s. The maximum allowed MS value is 9.5. The program uses by default 'IASPEI'.

- IASPEI** = Prague Formula from 1966 (see Willmore (1979) and citations therein)
- R-P** = formula 18 in Rezapour-Pearce (1998)

MS-ATTENUATION MODEL : IASPEI

For the calculation of local/regional ML from (regional/local) S- and Lg-type onsets one can choose between three models. The maximum allowed ML value is 7.5 and the default model is 'Bath':

Note: 'Richter' ML assumes original Wood-Anderson instrument or Wood-Anderson simulated amplitude measurements in [mm] and the period information is not used.

- | | |
|----------------|---|
| Model | file name (to be used for ML-CORRECTION FILE) |
| Bath | = MLCORR.TABLE
This period-depending attenuation relation (Båth <i>et al.</i> , 1976) is originally only defined for periods between 0.3 and 1.4 s. The program accepts period measurement values between 0.24 s (= 4.16 Hz) and 1.68 s (= 0.595 Hz). |
| Richter | = MLCORR.TABLE.wa
The original Wood-Anderson attenuation curve for California (Richter, 1935). |

Richter = *MLCORR.TABLE.wa.nicolas*

A Wood-Anderson type curve modified for Europe by Nicolas *et al.* (1982).

ML-ATTENUATION MODEL : Bath

Setting the file name (see above) defines the attenuation **ML-ATTENUATION MODEL**. The file must be either located in the run-directory, or the directory defined by the environment variable **HYPOSAT_DATA**, or one has to give the full file name including directory path information. By default, this is set to '*MLCORR.TABLE*'.

ML-CORRECTION FILE : MLCORR.TABLE

All station magnitude values calculated from the observations are used to calculate (mean) network magnitudes. These magnitudes are then listed in *hyposat-out* including their uncertainties.

If of interest, one can limit the distance range in [°] of contributing observations by setting one or more of the following parameters. By default, no distance limits (beside limits of the correction tables) are used.

MIN DISTANCE FOR MS : 0.0

MAX DISTANCE FOR MS : 180.0

MIN DISTANCE FOR MB : 0.0

MAX DISTANCE FOR MB : 180.0

MIN DISTANCE FOR ML : 0.0

MAX DISTANCE FOR ML : 20.0

By default, only one magnitude per station and magnitude type (*i.e.*, the largest one) is used to calculate a network magnitude. If setting the following parameter, all observed station magnitudes are used to calculate the network magnitude (*i.e.*, amplitude and period measurements *e.g.*, from both P and pP or sP onsets as long they are within a time window of 8 s after the 1st onset).

ALL STATION MAGNITUDES : 0

3.8 I/O RELATED PARAMETERS

3.8.1 Input

The default data-input file name is *hyposat-in*. Here, a different file name for the data-input file can be given. The default is also used when the parameter is set to ‘_’ or ‘’.

INPUT FILE NAME : *hyposat-in*

The *hyposat-in* format has slightly changed from **HYPOSAT 6** on. If the input file still follows the old syntax, one has to set this flag. The default is ‘0’ for using the new syntax.

HYPOSAT-IN OLD SYNTAX : 0

The final **HYPOSAT** solution can be compared with a reference hypocenter location and source time, which can be set with the following parameters. The default is that no reference location is used, ‘0’.

REFERENCE EVENT : 0

REFERENCE SOURCE LONGITUDE : -999.0

REFERENCE SOURCE LATITUDE : -999.0

REFERENCE SOURCE DEPTH : -999.0

3.8.2 Specific Parameters for ISF 1.0-formatted Input Files

Besides data input in **HYPOSAT** format, **HYPOSAT** can also read ISF (IMS) Bulletin formatted files as data input. The input-file name can be either *hyposat-in* or any other name as set with **INPUT FILE NAME**.

By default, the input is in **HYPOSAT** format as described in Chapter 2.

INPUT FORMAT ISF : 0

In the case of an ISF-formatted input file, the information about defining data (**TAS**) may be the result of an earlier location output. To force the program to use **all data** from the input file, one can set the following switch. The default setting is '0', which means using the **TAS** settings as defined by the input file.

ISF ALL PHASES : 0

ISF 1.0-formatted input has no given uncertainties (standard deviations) for the observations. Therefore, these values must be set by the program. To define standard deviations of the observed data, the reported quality of the onset readings is interpreted (see 'detection character' in the ISF/IMS documentation).

For higher precision onset time readings (input in milliseconds), the onset-time uncertainties set here are reduced by 20%. For non-P-type onset times, the values given here are doubled.

Standard deviation for i = impulsive onset readings (default 0.1 s):

ISF_i : 0.1

Standard deviation for e = emergent onset readings (default 0.5 s):

ISF_e : 0.5

Standard deviation for other, non-quality defined or q = questionable onset time readings (default 1 s):

ISF_o : 1.0

Standard deviation for BAZ observations (default 20°):

ISF_baz : 20.0

Standard deviation for slowness or ray parameter observations default (1 s/°):

ISF_slo : 1.0

In ISF-formatted files, the listed origins are identified with an author (source). Different authors are allowed but usually one location is identified as primary location. Here the author of the preferred reference location can be set. This location will then be used as reference location and as starting solution for the inversion process if **ISF EPICENTER** or **ISF DEPTH** are set.

If one wants to use the primary location, independently from the author, one has to give 'PRIME'. If the parameter is set to '_', just the 1st origin line in the input file is used. Here follow some examples:

ISF REFERENCE LOCATION : NAO

ISF REFERENCE LOCATION : PRIME

ISF REFERENCE LOCATION : ISC

ISF REFERENCE LOCATION : _

3.8.3 Output

One can set the author (*e.g.*, the analyst, the network name, the project name, *etc.*) of the event-location inversion: a free character string with maximum 10 characters. The parameter is set by default to 'HYPOSAT'.

AUTHOR OF SOLUTION : HYPOSAT

Program output is written by default in the file **hyposat-out**, which is also used when input is '_' or '. This name can be changed.

OUTPUT FILE NAME : hyposat-out

With this flag any output to a file can be switched off. By default, this flag is set to '1' and the output is written in the file defined by **OUTPUT FILE NAME**.

OUTPUT SWITCH : 1

HYPOSAT can additionally write out the last location results in an ISF-formatted file. The file name is either *hyposat-out.isf* or any other name defined by the parameter **OUTPUT FILE NAME** with the automatically added extension *.isf*. The option is set by the parameter **OUTPUT FORMAT ISF**, which is set by default to ('0'), no ISF-formatted output is written.

OUTPUT FORMAT ISF : 0

Any file name of the ISF formatted output file can be set with:

ISF OUTPUT FILE : file name

HYPOSAT can additionally write the location results in a JSON file. The file name is either *hyposat-out.json* or any other name defined by the parameter **OUTPUT FILE NAME** with the automatically added extension *.json*. The option is set by the parameter **OUTPUT FORMAT JSON**, which is set by default to ('0'), no JSON-formatted output is written.

OUTPUT FORMAT JSON : 0

Any file name of the JSON formatted output file can be set with:

JSON OUTPUT FILE : file name

The epicentral distance of the observations can be calculated in [km] or [°]. See also **MAX EPI DIST OF STAT** and **MIN EPI DIST OF STAT**.

If setting the flag **OUTPUT IN KM**, no ISF-formatted file can be written (**OUTPUT FORMAT ISF** is then automatically set to '0').

The default value '0' is for an epicentral-distance output in [°].

OUTPUT IN KM : 0

By default, all observations from stations laying outside the defined distance range (see **MIN EPI DIST OF STAT** and **MAX EPI DIST OF STAT**) are not listed in the **HYPOSAT** output files *hyposat-out* or *hyposat-out.isf*. If setting this flag to '1', the program lists all associated data (see the file *hyposat-in*) independently from the epicentral distance of the observing station. However, these data are then still not used to calculate the solution or any event location statistics (e.g., mean residuals, azimuthal gaps, etc.).

OUTPUT ALL DISTANCES : 0

HYPOSAT can calculate theoretical BAZ and slowness values for all observations and write these values in the *hyposat-out* file. If theoretical values are added to the output file, no residuals are calculated, and these fields stay empty. By default, no theoretical BAZ or slowness values are listed ('0'). This parameter setting is ignored in ISF formatted output files.

- 0** = no output of theoretical values
- 1** = theoretical BAZ are given in output
- 2** = theoretical slowness values are given in output
- 3** = both (BAZ & slowness values) are given in output

OUTPUT OF THEO. BAZ+P : 0

HYPOSAT can also calculate for all identified body-wave onsets the corresponding emergence angles at the seismic source. This flag is by default switched off.

FLAG EMERGENCE ANGLE OUTPUT : 0

If the ISF 1.0-formatted input file contains reported quality of the onset readings (see 'detection character' in the ISF/IMS documentation) such information will be also written out by default in *hyposat-out* and *hyposat-out.isf*. Based on the settings for **ISF_i**, **ISF_e** and **ISF_o** the applied data uncertainty during the last iteration can be translated into an uppercase 'detection character' and listed in the output files instead of the input from the ISF 1.0-formatted *hyposat-in*. The later also applies to non ISF 1.0-formatted *hyposat-in* input. Then, this flag must be set to '1'.

TT DATA UNCERTAINTY OUT : 0

HYPOSAT calculates two types of RMS values: one is the traditional RMS of all residuals from defining travel-time observations and the other value is the RMS calculated from the weighted residuals as done at the ISC. Both values are listed in the *hyposat-out* file. In the case of an additional ISF-formatted output file (see **OUTPUT FORMAT ISF**), the origin line contains only one place to write an RMS value. By default, the traditional RMS value is used. If the following switch is set to '1', the ISC-type RMS value is written in the ISF-formatted output file.

ISC-TYPE ISF RMS : 0

HYPOSAT can calculate an epicentral uncertainty ellipse based on the chosen **CONFIDENCE LEVEL** and the estimated hypocenter or epicenter and source time uncertainties. By default, this option is switched on '1'.

EPICENTER UNCERTAINTY ELLIPSE : 1

The primary and secondary azimuthal gaps (see *e.g.*, Bondár, 2011) and the Cyclic Polygon Quotient (CPQ) (Gallacher *et al.*, 2025) can be calculated for

- 0** = defining observations (default)
- 1** = all stations with reported observations

In any case, observations are considered only from stations in the allowed distance range (see **MAX EPI DIST OF STAT** and **MIN EPI DIST OF STAT**)

AZIMUTHAL GAP FOR OBSERVATIONS : 0

The applied local/regional model around the source can be listed at the end of the output file **hyposat-out** by setting this parameter to '1'. The default setting is no model output.

OUTPUT OF REGIONAL MODEL : 0

In ISF-formatted files, Event and Origin Identifications (IDs) can be defined.

Event ID: Any integer is allowed between 0 and 999999999. By default, it will be set to 0 (but then the number from an eventually used ISF-formatted input file is used).

ISF EVENT ID : 0

An ISF-formatted file can contain several event locations for the same event. The different locations can be separated by Origin IDs. The Origin ID can be any 8- character long string.

By default ('_') **HYPOSAT** uses, if it exists, the ID from the ISF-formatted input file.

ISF ORIGIN ID : _

Based on the location results and after eventually renaming and/or merging onsets, the program can write a new file in input format. The file name will be **hyposat-in.rev** or the name defined by **OUTPUT FILE NAME** with the extension **_rev**. By default, this option is switched off '0'.

FLAG NEW INPUT FILE : 0

HYPOSAT has different verbosity levels for output on the screen during program execution. This level can be defined here.

If **OUTPUT LEVEL** > 10, the output level for the screen is internally calculated and an extra output file **hyposat_gmi.out** is written with additional information about the Generalized Matrix Inversion (GMI) results. Then, the Resolution, Covariance, Correlation, and the Information- Density Matrix will be written out. A definition of the matrices can be found in *e.g.*, the textbooks of Aki & Richards (1980) or Menke (1989). The file **hyposat_gmi.out** always contains the named matrices for the last inversion. **OUTPUT LEVEL** can be set to the following values:

OUTPUT LEVEL	Output in hyposat_gmi.out	Output Level on Screen
any negative number	none	none
0 – 10	none	0 – 10
11	Resolution Matrix	4
12	Covariance Matrix	4
13	Correlation Matrix	4
14	All Three Matrices	4
15	"	5
16	"	6
17	"	7
18	"	8
19	"	9
20	", plus diagonal elements of the Information-Density Matrix	4
21 – 29	"	as for levels 11 – 19
30	", plus whole Information-Density Matrix	4
31 – 39	"	as for levels 11 – 19

As default value, the output (verbosity) level '4' is used.

OUTPUT LEVEL : 4

3.9 LIST OF OBSOLETE PARAMETERS AND THEIR NEW NAMES

Old Syntax		New Syntax
ATTENUATION MODEL	now	P-ATTENUATION MODEL
AUTOMATIC PROCESSING REGIONAL	now	AUTOMATIC PROCESSING
AUTOMATIC PROCESSING TELE	now	AUTOMATIC PROCESSING
AZIMUTH AS DEFINING	now	BAZ AS DEFINING
AZIMUTH ONLY INIT SOL	now	BAZ ONLY INIT SOL
AZIMUTH ONLY	now	BAZ ONLY
CRUST 1.0 PATH	now	ignored
CRUST 5.1	please see for	CRUST 1.0
DBLE SID. ERROR FACTOR	now	DBLE SID. UNCERTAINTY FACTOR
DOUBLE SIDED TIME ERROR	now	DOUBLE SIDED TIME UNCERTAINTY
DOWNWEIGHTING OF TASD	now	CHANGE WEIGHTING OF TASD
EPICENTRE ERROR ELLIPSE	now	EPICENTER UNCERTAINTY ELLIPSE
IS PHASE GROUP-VEL	now	IS-PHASE GROUP-VEL
MAX T RES FOR AZI OF B USE	now	MAX T RES FOR BAZ OF B USE
MAX T RES FOR AZI OF L USE	now	MAX T RES FOR BAZ OF L USE
MAXIMUM ALLOWED P RESIDUUM	now	MAXIMUM ALLOWED P RESIDUAL
MAXIMUM ALLOWED S RESIDUUM	now	MAXIMUM ALLOWED S RESIDUAL
MAXIMUM AZIMUTH ERROR	now	MAXIMUM BAZ RESIDUAL
MAXIMUM BAZ RESIDUUM	now	MAXIMUM BAZ RESIDUAL
MAXIMUM SLOWNESS ERROR	now	MAXIMUM SLOWNESS RESIDUAL
MAXIMUM SLOWNESS RESIDUUM	now	MAXIMUM SLOWNESS RESIDUAL
PLANE WAVE APPROX	see now for	NO PLANE WAVE APPROX
S-ATTENUATION MODEL	see now for	MS-ATTENUATION MODEL
STARTING DEPTH ERROR	now	STARTING DEPTH UNCERTAINTY
STARTING LATITUDE ERROR	now	STARTING LATITUDE UNCERTAINTY
STARTING LONGITUDE ERROR	now	STARTING LONGITUDE UNCERTAINTY
STARTING TIME ERROR	now	STARTING TIME UNCERTAINTY
T PHASE GROUP-VEL	now	T-PHASE GROUP-VEL
USE ALL AZIMUTH INFORMATION	now	USE ALL BAZ INFORMATION
# TO SEARCH OSCILLATIONS	now	ITERATIONS TO SEARCH OSCILLATIONS

4 THE *hyposat-out* FILE

When executing *../bin/hyposat* with the input files *hyposat-parameter.tele* and *hyposat-in.tele* located in the directory *./example/* one will get the following *hyposat-out* file: This example is the inversion result after running the inversion on a LINUX Centos7 system, on other systems the results might be slightly different due to numerical effects. Explanations are included in brackets []:

```
----- example for a hyposat-out file -----
HYPOSAT Version 6.3

Event solution by HYPOSAT

NORTHERN MOLUCCA SEA, 1996 29 June, converted from pIDC's Reviewed Event Bulletin (REB)

Initial epicenter calculated from 11 backazimuth-observation pairs
[ The BAZs used are mean values of all observations at one station. Not all possible backazimuth-observation pairs are used: if one station is more than
170 deg apart from the crossing point, this crossing point is skipped. ]

Parameters of initial solution (+/- 1 standard deviation):
Epicenter lat: 29.395 +/- 0.585 [deg]
Epicenter lon: 115.468 +/- 0.582 [deg]

[ type 1: S - P, type 2: Sg - Pg observation, type 3: Sn - Pn observation, type 4: Sb - Pb observation ]
S-P Travel-time difference type 1 with 2 observation(s)
to= 836008582.7 +/- 15.7 [s] Vp/Vs= 1.76 +/- 0.04
Mean source time: 836008582.700 +/- 15.671 [s]
Mean vp/vs: 1.758 +/- 0.035

Iterations : 9
Number of defining: 45
Reference model : ak135_A

The new source parameters:

Confidence level of given uncertainties: 68.27 %

Source time : 1996 06 29 00 36 42.439 +/- 0.089 [s]
or 836008602.439 +/- 0.089 [s]
or 1996-181:00.36.42.439 +/- 0.089 [s]

Epicenter lat: 1.4102 +/- 0.0164 [deg]
Epicenter lon: 126.3106 +/- 0.0345 [deg]
Source depth : 0.00 [km] Fixed by input

Epicenter uncertainty ellipse:
Major half axis: 4.19 [km] Minor half axis: 2.49 [km]
Azimuth: 142.1 [deg] Area: 32.78 [km^2]

Plinn-Engdahl Region ( 266 ) : Northern Molucca Sea

[ Here network magnitudes for MS and mb could be calculated including their standard deviations based on the single station-magnitude observations and
the chosen attenuation model(s) (see magnitude-parameter settings in hyposat-parameter). ]
Magnitude: 3.6 +/- 0.3 ( 2 obs, Ms, R-P)
Magnitude: 4.4 +/- 0.3 ( 14 obs, mb, G-R)

[ Stat - station name; Delta - epicentral distance in [deg] or [km]; Azi - azimuth from epicenter in [deg]; Phase - phase name from hyposat-in; [used]
- program chosen phase name (if not identical with Phase); Onset time - hours, minutes & seconds from hyposat-in; Res - travel time residual in [s]; Baz
- backazimuth (from station to epicenter) in [deg]; Res - Baz residual in [deg]; Rayp - ray parameter of onset in [s/deg]; Ray - ray parameter residual in
[s/deg]; Used - usage of input data as defining; SNR - signal-to-noise ratio; Amplitude - signal amplitude in [nm]; Period - signal period in [s]; MAG
- magnitude and magnitude type; Q - onset quality as defined by parameter settings in hyposat-parameter; Em-Ang - emergence angle of seismic onset at
the source in [deg]; ARID & Arrival-id ]
Stat Delta Azi Phase [used] Onset time Res Baz Res Rayp Res Used SNR Amplitude Period MAG Q Em-Ang ARID
WRA 22.635 160.04 P 00 41 44.700 -0.536 331.50 -7.23 11.10 0.48 TAS 97.80 4.50 0.300 4.44 mb E 33.63 abcdef01
WRA 22.635 160.04 S 00 45 48.700 -5.946 338.00 -0.73 17.00 -2.31 A 8.10 2.00 0.900 36.94 abcdef02
QIS 25.422 149.87 P1 P 00 42 12.800 0.734 T D 35.50 E 28.27 abcdef03
QIS 25.422 149.87 PcP 00 45 44.800 0.196 T D E 6.75 abcdef04
ASAR 25.997 163.95 P 00 42 16.900 -0.381 346.30 3.84 7.10 -1.96 TA D 35.10 3.40 0.500 4.23 mb E 28.19 abcdef05
ASAR 25.997 163.95 PcP 00 45 45.200 -0.691 345.10 2.64 2.30 0.00 TASD 11.50 2.20 0.500 E 6.88 abcdef06
ASAR 25.997 163.95 S 00 46 45.300 -3.898 347.60 5.14 20.30 4.49 A 6.10 3.90 0.800 29.48 abcdef07
WARB 27.434 179.35 P 00 42 31.200 1.012 339.40 -19.88 8.20 -0.78 T SD 8.20 6.50 0.700 4.51 mb E 27.91 abcdef08
WARB 27.434 179.35 PcP 00 45 49.400 0.213 T D E 7.20 abcdef09
MEEK 28.857 194.38 P 00 42 42.700 -0.205 T E 27.62 abcdef10
CMAR 31.745 303.97 P 00 43 9.000 0.450 109.70 -9.44 7.80 -1.00 TAS 4.30 0.60 0.400 3.88 mb E 27.31 abcdef11
[ The following BAZ observation of the LR onset was here not used, because its travel-time residual became too large. ]
CMAR 31.745 303.97 LR 00 57 48.700 26.463 110.00 -9.14 39.50 0.45 188.80 19.360 3.87 Ms E abcdef12
FORT 32.054 177.16 P 00 43 11.300 0.249 T 10.80 E 27.27 abcdef13
WOOL 32.603 187.39 P 00 43 15.600 -0.301 7.90 -0.72 9.90 1.14 TA 8.80 4.10 0.600 4.54 mb E 27.20 abcdef14
SHK 33.484 9.55 P 00 43 25.400 1.832 T 7.20 1.70 0.700 4.00 mb E 27.05 abcdef15
KSAR 35.885 2.14 Amb 00 43 46.700 2.592 T 7.20 40.10 19.860 3.24 Ms E 26.57 abcdef18
KSAR 35.885 2.14 P 00 43 46.700 2.438 177.30 -5.38 10.10 1.52 A 7.20 40.10 19.860 3.24 Ms E abcdef18
KSAR 35.885 2.14 AMs 01 04 25.000 E abcdef19
KSAR 35.885 2.14 LR 01 04 25.000 261.541 160.00 -22.68 46.00 6.96 E abcdef19
STKA 36.139 157.65 P 00 43 46.800 0.247 323.20 -10.26 9.00 0.44 T SD 8.30 7.20 0.600 4.67 mb E 26.52 abcdef16
STKA 36.139 157.65 PcP 00 46 12.000 -0.544 T D E 8.94 abcdef17
MJAR 36.651 16.15 P 00 43 51.900 1.001 357.10 156.90 18.80 10.27 T 2.00 0.550 4.10 mb E 26.41 abcdef20
PDY 59.022 351.98 P 00 46 47.000 3.081 111.20 -52.87 6.60 -0.34 S 23.80 4.30 0.450 4.78 mb E 21.22 abcdef21
ZAL 62.463 333.77 P 00 47 9.000 1.533 T 20.42 abcdef22
ABKT 72.023 309.48 P 00 48 9.500 0.890 T E 18.23 abcdef23
[ The Associated, but unnamed phase tx was identified as P onset and the corresponding residuals were calculated. ]
NRI 72.517 346.74 tx P 00 48 8.700 -2.083 195.90 56.33 3.90 -2.06 11.20 3.00 0.750 4.50 mb E 18.12 abcdef24
MAW 81.448 200.28 P 00 49 1.100 0.006 T E 16.04 abcdef25
KVAR 84.422 313.85 P 00 49 17.200 -0.116 T E 15.33 abcdef26
NPO 87.365 25.36 P 00 49 31.000 -0.273 T E 14.61 abcdef27
ARCES 92.466 339.77 P 00 49 51.700 -3.391 94.50 15.11 4.10 -0.52 S 10.50 0.80 0.550 4.31 mb E 13.94 abcdef28
SPITS 92.658 348.81 P 00 49 55.900 0.052 116.60 46.42 3.50 -1.12 T 10.40 3.60 0.900 4.77 mb E 13.93 abcdef29
FINES 93.664 331.72 P 00 50 0.000 -0.711 111.20 30.86 5.90 1.30 T 6.60 0.60 0.550 4.17 mb E 13.89 abcdef30
HFS 99.860 332.03 P 00 50 28.700 -0.136 311.30 -118.20 1.50 -2.95 T 3.70 1.10 0.750 4.58 mb E 13.41 abcdef31
SCHQ 122.908 9.02 PKPdf PKiP 00 55 41.200 0.263 T E 5.91 abcdef33
TXAR 123.329 53.16 PKPdf PKiP 00 55 42.700 0.040 T E 5.92 abcdef32
DBIC 130.605 279.96 PKPdf PKiP 00 55 57.100 -0.276 T E 6.03 abcdef34
PLCA 137.981 160.77 PKPdf 00 56 9.100 -0.689 311.90 106.23 6.40 4.56 T 4.40 0.90 0.900 E 5.51 abcdef35
LPZA 159.482 136.90 PKPdf 00 56 44.700 -0.492 T E 3.44 abcdef36

Defining travel-time differences:

Stat Delta Phases Observed Res
```

```
QIS      25.422 PcP - P      212.000 -0.538
ASAR     25.997 PcP - P      208.300 -0.310
WARB     27.434 PcP - P      198.200 -0.799
STKA     36.139 PcP - P      145.200 -0.791
```

Number of usable stations: 28

[The azimuth range of the maximum gap without any observations is always given in clockwise direction. For details about the location-accuracy Cyclic Polygon Quotient (CPQ), please see Gallacher et al., 2025.]
Maximum azimuthal gap of defining observations: 53.2 -> 136.9 [deg] = 83.7 [deg]; CPQ = 0.854

[The azimuth range of the maximum possible secondary gap (see Bondár et al., 2004) without any observations is always given in clockwise direction.]
Maximum secondary azimuthal gap of defining observations: 25.4 -> 136.9 [deg] = 111.5 [deg]; CPQ = 0.769

[RMS is defined as $\hat{\sigma}(\hat{\sigma}^2 \text{Res}^2 / N)$, MEAN-RES is defined as $(\hat{\sigma}^2 | \text{Res}|) / N$, and MEAN is defined as $(\hat{\sigma}^2 \text{Res}) / N$; all with the above listed residuals Res and the corresponding number of data N as listed in the first column.]

Residuals of defining data	RMS	MEAN-RES	MEAN
26 onset times	0.503	0.412	0.000 [s]
8 backazimuth values	5.238	4.390	-1.484 [deg]
7 ray parameters	0.586	0.508	-0.244 [s/deg]
4 travel-time differences	0.642	0.610	-0.610 [s]

[The weighted RMS as used at the ISC is defined as $\hat{\sigma}^2((\hat{\sigma}^2 \text{Res}^2 / w) / (\hat{\sigma}^2 l / w))$ with the above listed residuals Res and the data weights w used for the inversion (i.e., the applied standard deviations of the observed onset times)]
Weighted RMS of onset times (ISC type): 0.503 [s]

[The weighted misfit is here defined for the L1-norm as $(\hat{\sigma}^2 | \text{Res}| / w) / N$ and for the L2-norm as $\hat{\sigma}^2((\hat{\sigma}^2 (\text{Res}/w)^2) / N)$ with N the number of data as listed in the first column. Input data means here also data not used to locate the event. In this case, all backazimuth and ray parameter observations defined as usable by the switches in hyposat-in were also included.]

Weighted misfit of input data	L1	L2
33 onset times	2.818	4.179
18 backazimuth values	1.421	2.389
17 ray parameters	1.474	2.734
4 travel-time differences	1.437	1.514
72 misfit over all	2.075	3.365

[The following line is a repetition of the most important inversion results T0 â€” source time; LAT - source latitude; LON - source longitude; Z - source depth; VPVS - Vp / Vs as estimated by the Wadati approach; DLAT - latitude uncertainty; DLON - longitude uncertainty; DZ - depth uncertainty, if the depth was fixed, this is indicated here; DTO - source time uncertainty; DVPVS â€” VPVS uncertainty; DEF - number of defining data; RMS - root-mean-square of defining onset times]

T0	LAT	LON	Z	VPVS	DLAT	DLON	DZ	DTO	DVPVS	DEF	RMS
1996-06-29 00 36 42.439	1.410	126.311	0.00	1.76	0.0164	0.0345	Fixed	0.089	0.04	45	0.503

[However, we have still a fixed depth. Let us now try to fit the data better with another depth (see DEPTH FLAG is set to b!)]

Iterations : 3
Number of defining: 54
Reference model : ak135_A

The new source parameters:

Confidence level of given uncertainties: 68.27 %

Source time : 1996 06 29 00 36 49.185 +/- 0.496 [s]
or 836008609.185 +/- 0.496 [s]
or 1996-181:00:36.49.185 +/- 0.496 [s]

Epicenter lat: 1.3221 +/- 0.0152 [deg]
Epicenter lon: 126.2942 +/- 0.0360 [deg]
Source depth : 44.78 +/- 4.43 [km]

[The mean travel-time residual was not zero. The source time is therefore corrected for this bias.]
Source time corrected for mean travel-time residual (-0.032 [s])

Source time : 1996 06 29 00 36 49.153 +/- 0.496 [s]
or 836008609.153 +/- 0.496 [s]
or 1996-181:00:36.49.153 +/- 0.496 [s]

[Note the change of the epicenter uncertainty ellipse.]

Epicenter uncertainty ellipse:
Major half axis: 8.63 [km] Minor half axis: 1.88 [km]
Azimuth: 45.7 [deg] Area: 51.06 [km²]

Flinn-Engdahl Region (266): Northern Molucca Sea

Magnitude: 3.6 +/- 0.3 (2 obs, Ms, R-P)
Magnitude: 4.4 +/- 0.3 (14 obs, mb, G-R)

Stat	Delta	Azi	Phase	[used]	Onset time	Res	Baz	Res	Rayp	Res	Used	SNR	Amplitude	Period	MAG	Q	Em-Ang	ARID
WRA	22.558	159.93	P		00 41 44.700	-0.978	331.50	-7.11	11.10	0.51	TASD	97.80	4.50	0.300	4.36	mb	E	50.24 abcdef01
WRA	22.558	159.93	S		00 45 48.700	-1.882	338.00	-0.61	17.00	0.65	TASD	8.10	2.00	0.900			E	41.39 abcdef02
QIS	25.355	149.75	P1	P	00 42 12.800	0.543					T D	35.50					E	41.16 abcdef03
QIS	25.355	149.75	PcP		00 45 44.800	0.556					T D						E	9.38 abcdef04
ASAR	25.918	163.87	P		00 42 16.900	-0.457	346.30	3.93	7.10	-1.95	TA D	35.10	3.40	0.500	4.19	mb	E	41.02 abcdef05
ASAR	25.918	163.87	PcP		00 45 45.200	-0.303	345.10	2.73	2.30	0.00	TASD	11.50	2.20	0.500			E	9.56 abcdef06
ASAR	25.918	163.87	S		00 46 45.300	0.622	347.60	5.23	20.30	4.49	TA D	6.10	3.90	0.800			E	39.71 abcdef07
WARB	27.347	179.32	P		00 42 31.200	1.020	339.40	-19.84	8.20	-0.76	T SD	8.20	6.50	0.700	4.41	mb	E	40.54 abcdef08
WARB	27.347	179.32	PcP		00 45 49.400	0.622					T D						E	10.00 abcdef09
MEBK	28.768	194.39	P		00 42 42.700	-0.169						6.80					E	40.11 abcdef10
CMAR	31.781	304.10	P		00 43 9.000	-0.591	109.70	-9.59	7.80	-0.99	TAS	4.30	0.60	0.400	3.79	mb	E	39.58 abcdef11
CMAR	31.781	304.10	LR		00 57 48.700	18.366	110.00	-9.29	39.50	0.45		10.80	188.80	19.360	3.87	Ms	E	abcdef12
FORT	31.968	177.13	P		00 43 11.300	0.282					T	4.80					E	39.54 abcdef13
WOOL	32.514	187.38	P		00 43 15.600	-0.245	7.90	-0.71	9.90	1.14	TA	8.80	4.10	0.600	4.47	mb	E	39.42 abcdef14
SHK	33.573	9.55	P		00 43 25.400	0.349					T						E	39.12 abcdef15
KSAR	35.973	2.15	Amb		00 43 46.700	1.153						7.20	1.70	0.700	4.03	mb	E	abcdef18
KSAR	35.973	2.15	P		00 43 46.700	1.012	177.30	-5.41	10.10	1.54	TA	7.20					E	38.34 abcdef18
KSAR	35.973	2.15	AMS		01 04 25.000								40.10	19.860	3.24	Ms	E	abcdef19
KSAR	35.973	2.15	LR		01 04 25.000	251.389	160.00	-22.71	46.00	6.96							E	abcdef19
STKA	36.065	157.58	P		00 43 46.800	0.214	323.20	-10.18	9.00	0.45	T SD	8.30	7.20	0.600	4.72	mb	E	38.31 abcdef16
STKA	36.065	157.58	PcP		00 46 12.000	-0.160					T D						E	12.44 abcdef17
MJAR	36.740	16.14	P		00 43 51.900	-0.416	357.10	156.92	18.80	10.29	T		2.00	0.550	4.17	mb	E	38.09 abcdef20
PDY	59.107	352.00	P		00 46 47.000	2.168	111.20	-52.91	6.60	-0.32	S	23.80	4.30	0.450	4.79	mb	E	30.09 abcdef21
ZAL	62.534	333.80	P		00 47 9.000	0.772					T						E	28.90 abcdef22
ABKT	72.066	309.51	P		00 48 9.500	0.459					T						E	25.68 abcdef23
NRI	72.599	346.76	tx	P	00 48 8.700	-2.733	195.90	56.30	3.90	-2.04		11.20	3.00	0.750	4.37	mb	E	25.49 abcdef24
MAW	81.360	200.28	P		00 49 1.100	0.397					T						E	22.54 abcdef25
KVAR	84.471	313.86	P		00 49 17.200	-0.406					T						E	21.46 abcdef26
NPO	87.451	25.36	P		00 49 31.000	-0.701											E	20.36 abcdef27
ARCES	92.543	339.77	P		00 49 51.700	-3.732	94.50	15.07	4.10	-0.52	S	10.50	0.80	0.550	4.29	mb	E	19.52 abcdef28
SPITS	92.741	348.81	P		00 49 55.900	-0.317	116.60	46.38	3.50	-1.11	T	10.40	3.60	0.900	4.74	mb	E	19.50 abcdef29
FINES	93.733	331.71	P		00 50 0.000	-1.017	111.20	30.81	5.90	1.30	T	6.60	0.60	0.550	4.19	mb	E	19.44 abcdef30
HFS	99.930	332.02	P	Pdif	00 50 28.700	-0.416	311.30	-118.25	1.50	-2.95	T	3.70	1.10	0.750	4.57	mb	E	18.77 abcdef31
SCHQ	122.997	9.01	PKPdf		00 55 41.200	0.524					T						E	7.98 abcdef33
TXAR	123.395	53.20	PKPdf		00 55 42.700	0.371					T						E	7.97 abcdef32
DBIC	130.604	279.88	PKPdf		00 55 57.100	0.812					T						E	7.88 abcdef34
PLCA	137.904	160.82	PKPdf		00 56 9.100	-0.319	311.90	106.30	6.40	4.56	T	4.40	0.90	0.900			E	7.65 abcdef35
LPZA	159.429	137.09	PKPdf		00 56 44.700	-0.178					T						E	4.78 abcdef36

Defining travel-time differences:

Stat	Delta	Phases	Observed	Res
------	-------	--------	----------	-----

```

WRA 22.558 S - P      244.000 -0.904
QIS 25.355 PcP - P    212.000  0.013
ASAR 25.918 PcP - P   208.300  0.154
ASAR 25.918 S - P     268.400  1.079
ASAR 25.918 S - PcP    60.100  0.925
WARB 27.347 PcP - P    198.200 -0.398
STKA 36.065 PcP - P    145.200 -0.374

```

Number of usable stations: 28

[Here we get the number of all iterations e.g., also including the earlier solution for fixed depth.]
Total number of iterations: 21

Maximum azimuthal gap of defining observations: 53.2 -> 137.1 [deg] = 83.9 [deg]; CPQ = 0.853

Maximum secondary azimuthal gap of defining observations: 25.4 -> 137.1 [deg] = 111.7 [deg]; CPQ = 0.769

```

Residuals of defining data      RMS      MEAN-RES      MEAN
31 onset times      :      0.654      0.552      0.000 [s]
8 backazimuth values :      5.279      4.416     -1.441 [deg]
8 ray parameters    :      0.592      0.525     -0.121 [s/deg]
7 travel-time differences :      0.672      0.550      0.071 [s]

```

Weighted RMS of onset times (ISC type): 0.614 [s]

```

Weighted misfit of input data      L1      L2
33 onset times      :      2.198      3.122
18 backazimuth values :      1.421      2.389
17 ray parameters    :      1.428      2.726
7 travel-time differences :      0.935      1.075
75 misfit over all      :      1.719      2.729

```

```

T0      LAT      LON      Z      VPVS      DLAT      DLON      DZ      DT0      DVPVS DEF      RMS
1996-06-29 00 36 49.153  1.322 126.294  44.78  1.76  0.0152  0.0360  4.43  0.496  0.04  54  0.654

```

----- end of the *hyposat-out* example file -----

5 THE PROGRAM HYPOMOD AND THE *hypomod-out* FILE

In some cases, it may be of interest to know how well a set of observations fits with a specific source location. The program **HYPOMOD** calculates for a given seismic hypocenter solution the residuals for all observed data: the travel times, travel-time differences, the backazimuths and the slowness values. With the example given here for a hypocenter inversion with **HYPOSAT**, one has only to modify slightly the *hyposat-parameter* file and then one can execute the program **HYPOMOD**. The only modifications needed in *hyposat-parameter* are to set as starting source parameters the inversion results. Then, ignoring all inversion related settings in *hyposat-parameter* one will get an output-file called by default *hypomod-out*. This default name can be modified by setting the parameter **OUTPUT FILE NAME** in *hyposat-parameter* different than the default *hyposat-out*. The file *hypomod-out* has in principle the same format as *hyposat-out*.

HYPOMOD calculates residuals and the statistical parameters for this solution from **all(!)** input data as defined in the *hypomod-in* input file (default name), which has the same contents and format as *hyposat-in*. Therefore, the statistical parameters as *e.g.*, mean residuals or RMS values are different from the ones calculated after the final inversion in **HYPOSAT** (please compare with the 2nd solution in the file *hyposat-out.tele*).

No output can be written in ISF or JSON format.

----- example for changes in the *hyposat-parameter* file -----

The following *hyposat-parameter* flag is used only with **HYPOMOD**. It allows to search for the best fitting phase name for an associated body wave independently from the name given in the input file. By default, this option is switched off ('0').

FLAG FREE PHASE SEARCH : 0

Then, one has also to give the source parameters. Here, the location results from the teleseismic event example in the directory *./examples/* are used (see also Chapter 4), so the file *hypomod-in* is identical with the file *hyposat-in* of Chapter 2 and which was used for the example in Chapter 4. In this example the parameter **FLAG FREE PHASE SEARCH** was set to '1'.

INPUT FILE NAME : *hypomod-in*

OUTPUT FILE NAME : *hypomod-out*

STARTING SOURCE LATITUDE : 1.3221

STARTING SOURCE LONGITUDE : 126.2942

STARTING SOURCE DEPTH : 44.78

STARTING SOURCE TIME : 836008609.155

or

STARTING SOURCE TIME : 1996-181:00.36.49.155

or

STARTING SOURCE TIME : 1996-06-29 00:36:49.155

When running **HYPOMOD** it can also be of interest to ask for the theoretical BAZs and ray parameters (see Chapter 3 for details) from all observations (as also done in our example):

OUTPUT OF THEO. BAZ+P : 3

----- end of example -----

When executing on the Linux system the C-shell script **./run_hypomod_example** in the directory **./examples/**, you should get an output file **hypomod-out** looking like this, with explanations included again in brackets []:

----- example for a **hypomod-out** file -----

```
HYPOMOD Version 2.3

Event solution by input from HYPOSAT 6.3

NORTHERN MOLUCCA SEA, 1996 29 June, converted from pIDC's Reviewed Event Bulletin (REB)

Reference model : akl35_A

The source parameters:

Source time : 1996 06 29 00 36 49.153
or           836008609.153 [s]
or           1996-181:00.36.49.153

Epicenter lat: 1.3221 [deg]
Epicenter lon: 126.2942 [deg]
Source depth : 44.78 [km]

Flinn-Engdahl Region ( 266 ): Northern Molucca Sea

Magnitude: 3.6 +/- 0.3 ( 2 obs, MS, R-P)
Magnitude: 4.4 +/- 0.3 ( 14 obs, mb, G-R)

Stat Delta Azi Phase [used] Onset time Res Baz Res Rayp Res Used SNR Amplitude Period MAG Q Em-Ang ARID
WRA 22.558 159.93 P 00 41 44.700 -0.978 331.50 -7.11 11.10 0.51 TASD 97.80 4.50 0.300 4.36 mb - 50.24 abcdef01
WRA 22.558 159.93 S 00 45 48.700 -1.882 338.00 -0.61 17.00 0.65 TASD 8.10 2.00 0.900 - 41.39 abcdef02
[ For the following phase, no BAZ or ray parameter observations were reported. Because of the setting for OUTPUT OF THEO. BAZ+P to '3', the program fills in the theoretical values without calculating any residuals. ]
QIS 25.355 149.75 P 00 42 12.800 0.544 327.49 9.07 T D 35.50 - 41.16 abcdef03
QIS 25.355 149.75 PcP 00 45 44.800 0.556 327.49 2.25 T D - 9.38 abcdef04
ASAR 25.918 163.87 P 00 42 16.900 -0.456 346.30 3.93 7.10 -1.95 TASD 35.10 3.40 0.500 4.19 mb - 41.02 abcdef05
ASAR 25.918 163.87 PcP 00 45 45.200 -0.303 345.10 2.73 2.30 0.00 TASD 11.50 2.20 0.500 - 9.56 abcdef06
ASAR 25.918 163.87 S 00 46 45.300 0.623 347.60 5.23 20.30 4.49 TASD 6.10 3.90 0.800 - 39.71 abcdef07
WARB 27.347 179.32 P 00 42 31.200 1.021 339.40 -19.84 8.20 -0.76 TASD 8.20 6.50 0.700 4.41 mb - 40.54 abcdef08
WARB 27.347 179.32 PcP 00 45 49.400 0.622 359.24 2.40 T D - 10.00 abcdef09
MEEK 28.768 194.39 P 00 42 42.700 -0.169 16.12 8.88 T - 40.11 abcdef10
CMAR 31.781 304.10 P 00 43 9.000 -0.591 109.70 -9.59 7.80 -0.99 TAS 4.30 0.60 0.400 3.79 mb - 39.58 abcdef11
CMAR 31.781 304.10 LR 00 57 48.700 18.367 110.00 -9.29 39.50 0.45 A 188.80 19.360 3.87 MS - abcdef12
FORT 31.968 177.13 P 00 43 11.300 0.283 356.67 8.78 T - 39.54 abcdef13
WOOL 32.514 187.38 P 00 43 15.600 -0.244 7.90 -0.71 9.90 1.14 TAS 8.80 4.10 0.600 4.47 mb - 39.42 abcdef14
SHK 33.573 9.55 P 00 43 25.400 0.349 191.60 8.70 T - 39.12 abcdef15
KSAR 35.973 2.15 P 00 43 46.700 1.012 177.30 -5.41 10.10 1.54 TAS 7.20 1.70 0.700 4.03 mb - 38.34 abcdef18
KSAR 35.973 2.15 LR 01 04 25.000 251.389 160.00 -22.71 46.00 6.96 A 40.10 19.860 3.24 MS - abcdef19
STKA 36.065 157.58 P 00 43 46.800 0.214 323.20 -10.18 9.00 0.45 TASD 8.30 7.20 0.600 4.72 mb - 38.31 abcdef16
STKA 36.065 157.58 PcP 00 46 12.000 -0.160 333.38 2.98 T D - 12.44 abcdef17
MJAR 36.740 16.14 P 00 43 51.900 -0.416 357.10 156.92 18.80 10.29 TAS 2.00 0.550 4.17 mb - 38.09 abcdef20
PDY 59.107 352.00 P 00 46 47.000 2.168 111.20 -52.91 6.60 -0.32 TAS 23.80 4.30 0.450 4.79 mb - 30.09 abcdef21
ZAL 62.534 333.80 P 00 47 9.000 0.773 131.71 6.67 T - 28.90 abcdef22
ABKT 72.066 309.51 P 00 48 9.500 0.459 102.71 5.98 T - 25.68 abcdef23
NRI 72.599 346.76 tx P 00 48 8.700 -2.733 195.90 56.30 3.90 -2.04 A 11.20 3.00 0.750 4.37 mb - 25.49 abcdef24
MAW 81.360 200.28 P 00 49 1.100 0.398 64.74 5.29 T - 22.54 abcdef25
KVAR 84.471 313.86 P 00 49 17.200 -0.406 93.49 5.05 T - 21.46 abcdef26
NPO 87.451 25.36 P 00 49 31.000 -0.701 267.69 4.81 T - 20.36 abcdef27
ARCES 92.543 339.77 P 00 49 51.700 -3.732 94.50 15.07 4.10 -0.52 TAS 10.50 0.80 0.550 4.29 mb - 19.52 abcdef28
SPITS 92.741 348.81 P 00 49 55.900 -0.317 116.60 46.38 3.50 -1.11 TAS 10.40 3.60 0.900 4.74 mb - 19.50 abcdef29
FINES 93.733 331.71 P 00 50 0.000 -1.017 111.20 30.81 5.90 1.30 TAS 6.60 0.60 0.550 4.19 mb - 19.44 abcdef30
HFS 99.930 332.02 P Pdif 00 50 28.700 -0.416 311.30 -118.25 1.50 -2.95 TAS 3.70 1.10 0.750 4.57 mb - 18.77 abcdef31
[ Because the parameter FLAG FREE PHASE SEARCH was set to '1', the program identifies the onsets as PkIKP, which has a smaller travel-time residual ]
SCHQ 122.997 9.01 PKPdif PKIKP 00 55 41.200 0.310 344.29 1.97 T - 8.21 abodef33
TXAR 123.395 53.20 PKPdif PKIKP 00 55 42.700 0.133 293.54 1.98 T - 8.22 abcdef32
DBIC 130.604 279.88 PKPdif PKIKP 00 55 57.100 -0.054 82.53 2.01 T - 8.37 abcdef34
PLCA 137.904 160.82 PKPdif 00 56 9.100 -0.319 311.90 106.30 6.40 4.56 TAS 4.40 0.90 0.900 - 7.65 abcdef35
LPZA 159.429 137.09 PKPdif 00 56 44.700 -0.178 225.14 1.15 T - 4.78 abcdef36

Defining travel-time differences:

Stat Delta Phases Observed Res
WRA 22.558 S - P 244.000 -0.904
QIS 25.355 PcP - P 212.000 0.013
ASAR 25.918 PcP - P 208.300 0.154
```

```

ASAR  25.918 S - P          268.400  1.079
ASAR  25.918 S - PcP        60.100  0.925
WARB  27.347 PcP - P        198.200 -0.399
STKA  36.065 PcP - P        145.200 -0.374

```

Number of usable stations: 28

Maximum azimuthal gap of all observations: 53.2 -> 137.1 [deg] = 83.9 [deg] CPQ = 0.854

Maximum secondary azimuthal gap of all observations: 25.4 -> 137.1 [deg] = 111.7 [deg] CPQ = 0.769

```

Residuals of defining data      RMS      MEAN-RES      MEAN
33 onset times      :      0.968      0.661     -0.087 [s]
20 backazimuth values :    55.011     34.014     8.353 [deg]
17 ray parameters    :      3.151      1.973     0.962 [s/deg]
 7 travel-time differences :      0.672      0.550     0.071 [s]

```

```

T0          LAT      LON      Z      VPVS      DLAT      DLON      DZ      DT0      DVPVS DEF      RMS
1996-06-29 00 36 49.153  1.322 126.294  44.78  0.00  0.0000  0.0000 Fixed  0.000  0.00  77  0.968

```

----- end of the ***hypomod-out*** file example-----

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